

附录 B 实验详细数据

B.1 精确求解器 (LB_2) 结果对比

表 B.1 精确求解器 (UBHMM + LB_2 + BFS) 实验结果对比

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-3-1	0.00068	242.473	79	14	6	1	0.000766993	242.473	108	26	0	1
3-3-2	0.00040	211.687	38	5	4	1	0.000439167	211.687	55	12	0	1
3-3-3	0.00018	190.653	14	3	0	1	0.000172138	190.653	14	3	0	1
3-3-4	0.00047	226.621	47	7	4	1	0.000526428	226.621	71	16	0	1
3-3-5	0.00112	262.589	130	21	15	1	0.002068281	262.589	321	85	0	1
3-3-6	0.00072	226.315	76	16	3	1	0.000742197	226.315	93	23	0	1
3-3-7	0.00095	242.167	114	27	7	1	0.001178503	242.167	159	45	0	1
3-3-8	0.00132	262.283	161	37	12	1	0.001603127	262.283	246	71	0	1
3-3-9	0.00057	211.381	67	17	3	1	0.000562668	211.381	76	22	0	1
3-3-10	0.00024	191.265	15	1	1	1	0.000240803	191.265	24	3	0	1
3-3-11	0.00103	226.315	114	23	8	1	0.001369715	226.315	181	45	0	1
3-3-12	0.00049	211.075	44	5	5	1	0.000619411	211.075	83	19	0	1
3-3-13	0.00070	236.985	71	6	14	1	0.001397133	236.985	176	46	0	1
3-3-14	0.00092	241.555	106	24	8	1	0.001090527	241.555	153	43	0	1
3-3-15	0.00033	170.231	28	7	0	1	0.000322104	170.231	28	7	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-3-16	0.00059	200.711	53	11	3	1	0.000628471	200.711	68	17	0	1
3-3-17	0.00092	226.315	79	13	11	1	0.001142502	226.315	176	51	0	1
3-3-18	0.00044	216.869	37	8	1	1	0.000351906	216.869	41	10	0	1
3-3-19	0.00027	190.959	20	4	1	1	0.000269175	190.959	25	6	0	1
3-3-20	0.00072	237.597	81	18	7	1	0.000775099	237.597	111	31	0	1
3-3-21	0.00061	205.893	65	12	5	1	0.000822544	205.893	111	28	0	1
3-3-22	0.00059	221.439	59	12	3	1	0.000620365	221.439	79	19	0	1
3-3-23	0.00019	175.719	13	3	0	1	0.000225782	175.719	13	3	0	1
3-3-24	0.00049	226.315	53	7	6	1	0.000965118	226.315	156	39	0	1
3-3-25	0.00044	221.439	42	7	5	1	0.000551939	221.439	70	19	0	1
3-3-26	0.00020	175.719	15	3	1	1	0.000237942	175.719	19	5	0	1
3-3-27	0.00029	180.595	21	4	1	1	0.00026679	180.595	25	6	0	1
3-3-28	0.00068	237.291	74	17	3	1	0.0006845	237.291	95	25	0	1
3-3-29	0.00046	222.051	41	4	6	1	0.00067234	222.051	81	18	0	1
3-3-30	0.00105	257.713	115	21	9	1	0.001250267	257.713	185	46	0	1
3-3-31	0.00045	195.835	40	5	6	1	0.000528336	195.835	66	17	0	1
3-3-32	0.00041	201.323	38	8	1	1	0.000383615	201.323	46	11	0	1
3-3-33	0.00006	129.387	1	0	0	1	4.76837E-05	129.387	1	0	0	1
3-3-34	0.00031	169.925	27	6	1	1	0.000268698	169.925	28	7	0	1
3-3-35	0.00076	206.199	95	20	8	1	0.001031876	206.199	160	45	0	1
3-3-36	0.00014	160.173	8	1	0	1	0.000176191	160.173	8	1	0	1
3-3-37	0.00050	200.711	51	10	3	1	0.000649452	200.711	72	17	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-3-38	0.00061	221.745	68	11	9	1	0.000762463	221.745	111	30	0	1
3-3-39	0.00016	149.809	11	2	0	1	0.000150681	149.809	11	2	0	1
3-3-40	0.00091	262.283	97	19	6	1	0.001015902	262.283	140	35	0	1
3-4-1	0.00083	253.592	93	23	9	1	0.000936985	253.592	136	44	0	1
3-4-2	0.01563	371.554	2094	587	240	1	0.044721603	371.554	7067	2561	0	1
3-4-3	0.00282	330.404	371	91	46	1	0.008253336	330.404	1352	473	0	1
3-4-4	0.00452	309.676	617	175	73	1	0.010741949	309.676	1747	655	0	1
3-4-5	0.00054	264.874	57	16	2	1	0.000539064	264.874	64	20	0	1
3-4-6	0.00278	320.958	333	91	33	1	0.003537655	320.958	517	181	0	1
3-4-7	0.00068	294.13	73	16	6	1	0.00126195	294.13	186	54	0	1
3-4-8	0.00613	330.404	842	268	52	1	0.019066095	330.404	3110	1153	0	1
3-4-9	0.00116	273.402	132	22	28	1	0.003029823	273.402	450	150	0	1
3-4-10	0.00525	325.528	655	155	94	1	0.012088776	325.528	1848	644	0	1
3-4-11	0.00128	325.834	148	38	12	1	0.00218153	325.834	325	106	0	1
3-4-12	0.00043	268.832	38	9	1	1	0.000435114	268.832	49	12	0	1
3-4-13	0.00158	299.312	172	25	39	1	0.005610943	299.312	869	282	0	1
3-4-14	0.00141	294.742	181	56	11	1	0.00175643	294.742	270	95	0	1
3-4-15	0.00091	284.072	92	12	16	1	0.001856565	284.072	272	78	0	1
3-4-16	0.00045	258.774	42	8	4	1	0.000509977	258.774	66	20	0	1
3-4-17	0.00220	304.8	294	77	35	1	0.006554365	304.8	1076	375	0	1
3-4-18	0.00423	309.982	526	143	62	1	0.008151054	309.982	1227	438	0	1
3-4-19	0.00050	248.41	54	13	5	1	0.000483036	248.41	59	18	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-4-20	0.00023	228.6	18	4	0	1	0.000192642	228.6	18	4	0	1
3-4-21	0.00863	330.404	1084	261	155	1	0.01863265	330.404	2702	938	0	1
3-4-22	0.00465	340.462	590	127	72	1	0.021635771	340.462	3668	1213	0	1
3-4-23	0.00151	294.742	173	28	28	1	0.00299859	294.742	479	152	0	1
3-4-24	0.00292	299.312	332	71	45	1	0.004693031	299.312	687	219	0	1
3-4-25	0.00312	284.378	411	107	51	1	0.009514332	284.378	1494	550	0	1
3-4-26	0.00239	273.402	268	81	23	1	0.002930641	273.402	403	146	0	1
3-4-27	0.01969	351.132	2736	741	372	1	0.120882988	351.132	17347	6359	0	1
3-4-28	0.00206	274.32	251	49	43	1	0.006421089	274.32	1014	338	0	1
3-4-29	0.00172	320.958	205	41	37	1	0.003362417	320.958	489	152	0	1
3-4-30	0.00989	320.04	1325	308	234	1	0.02957058	320.04	4655	1652	0	1
3-4-31	0.00043	253.592	48	10	6	1	0.000500917	253.592	73	22	0	1
3-4-32	0.00153	268.832	199	54	22	1	0.001705647	268.832	267	94	0	1
3-4-33	0.00232	304.8	269	56	46	1	0.005856037	304.8	841	279	0	1
3-4-34	0.00129	263.956	140	36	16	1	0.002171755	263.956	300	104	0	1
3-4-35	0.00329	289.56	438	121	45	1	0.006823301	289.56	1150	410	0	1
3-4-36	0.00108	259.386	123	25	17	1	0.001842022	259.386	256	82	0	1
3-4-37	0.00218	325.528	261	56	47	1	0.005601406	325.528	834	289	0	1
3-4-38	0.01167	330.098	1540	479	112	1	0.023825407	330.098	3710	1363	0	1
3-4-39	0.00252	294.436	341	76	62	1	0.004082918	294.436	688	240	0	1
3-4-40	0.00215	315.776	239	53	32	1	0.002761841	315.776	370	118	0	1
3-5-1	0.00423	375.053	494	109	87	1	0.012816906	375.053	1892	687	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-5-2	0.00115	303.117	120	34	11	1	0.001189947	303.117	151	51	0	1
3-5-3	0.08854	451.253	10443	3278	1100	1	0.208389759	451.253	25348	9709	0	1
3-5-4	0.00751	432.055	874	244	125	1	0.012901783	432.055	1694	629	0	1
3-5-5	0.15590	442.113	17637	4738	3171	1	0.719961882	442.113	81241	31470	0	1
3-5-6	0.00371	411.327	418	94	69	1	0.008047819	411.327	1093	372	0	1
3-5-7	0.00275	426.567	319	78	41	1	0.009993553	426.567	1410	509	0	1
3-5-8	0.01244	411.021	1518	403	237	1	0.128192663	411.021	16496	6502	0	1
3-5-9	0.00605	411.327	668	144	131	1	0.026921749	411.327	3733	1310	0	1
3-5-10	0.00093	323.845	94	29	4	1	0.000925779	323.845	108	36	0	1
3-5-11	0.00419	410.715	521	130	82	1	0.016079903	410.715	2506	939	0	1
3-5-12	0.02299	384.499	2644	819	247	1	0.071576118	384.499	10311	4060	0	1
3-5-13	0.00261	374.441	269	69	30	1	0.004490614	374.441	566	189	0	1
3-5-14	0.03310	396.393	3770	475	1241	1	0.583099604	396.393	63605	23181	0	1
3-5-15	0.01186	380.541	1474	443	189	1	0.026498079	380.541	3668	1442	0	1
3-5-16	0.00780	384.805	980	256	166	1	0.023741007	384.805	3360	1275	0	1
3-5-17	0.04205	415.897	5215	1606	730	1	0.35858202	415.897	43114	17452	0	1
3-5-18	0.00118	359.507	135	44	7	1	0.001126051	359.507	142	51	0	1
3-5-19	0.02387	401.269	3015	882	419	1	0.103530645	401.269	13896	5467	0	1
3-5-20	0.00295	410.103	330	109	18	1	0.003429651	410.103	457	166	0	1
3-5-21	0.00262	389.681	317	56	73	1	0.037706852	389.681	6175	2226	0	1
3-5-22	0.01909	400.963	2254	724	198	1	0.028429508	400.963	3742	1426	0	1
3-5-23	0.00761	406.451	873	201	119	1	0.039093971	406.451	5747	2090	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-5-24	0.00461	339.391	565	160	70	1	0.011321545	339.391	1603	606	0	1
3-5-25	0.05221	421.691	6579	1926	809	1	0.278039694	421.691	36137	14099	0	1
3-5-26	0.02029	410.715	2211	385	517	1	0.085489035	410.715	11413	4089	0	1
3-5-27	0.00347	380.235	394	126	30	1	0.005233765	380.235	684	253	0	1
3-5-28	0.00646	375.053	753	228	105	1	0.009818792	375.053	1265	499	0	1
3-5-29	0.00216	370.177	241	39	51	1	0.015210152	370.177	2357	813	0	1
3-5-30	0.00548	385.723	661	220	59	1	0.010076523	385.723	1384	555	0	1
3-5-31	0.36894	513.437	39267	10884	5831	1	2.809346914	513.437	257228	98605	0	1
3-5-32	0.00078	328.721	61	12	6	1	0.001740932	328.721	216	69	0	1
3-5-33	0.00094	333.903	92	29	4	1	0.0010221	333.903	119	42	0	1
3-5-34	0.00973	400.351	1100	177	306	1	0.043891907	400.351	5938	2179	0	1
3-5-35	0.00492	349.143	580	159	72	1	0.011995792	349.143	1765	645	0	1
3-5-36	0.03540	440.889	4439	1418	495	1	0.094722271	440.889	13115	5197	0	1
3-5-37	0.00564	411.021	695	169	123	1	0.035439014	411.021	5229	1965	0	1
3-5-38	0.00755	369.565	876	246	145	1	0.012650251	369.565	1576	605	0	1
3-5-39	0.12010	473.511	13973	4116	1740	1	0.6080935	473.511	65087	24823	0	1
3-5-40	0.04347	405.839	5642	1580	909	1	0.6090765	405.839	71513	28889	0	1
3-6-1	1.16856	549.242	108804	38495	12234	1	15.31521964	549.242	1169946	498457	0	1
3-6-2	0.01241	463.596	1304	282	271	1	0.063320398	463.596	8180	3151	0	1
3-6-3	0.00069	438.298	61	19	1	1	0.000621319	438.298	62	20	0	1
3-6-4	0.10078	519.986	11553	3182	2075	1	5.378494024	519.986	455525	187440	0	1
3-6-5	0.11344	545.896	11401	2687	2518	1	0.699852467	545.896	64067	24651	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-6-6	0.02787	503.828	3235	855	665	1	0.442937374	503.828	48350	19913	0	1
3-6-7	0.18533	498.952	19746	6256	3169	1	1.526916265	498.952	134718	56348	0	1
3-6-8	0.26178	514.192	27758	9183	3617	1	1.331199169	514.192	127155	53208	0	1
3-6-9	0.16029	581.558	15723	4823	2371	1	0.622891903	581.558	60299	24061	0	1
3-6-10	0.00246	462.678	279	80	36	1	0.011027336	462.678	1557	604	0	1
3-6-11	0.06617	519.374	7431	2746	563	1	0.166495085	519.374	18727	7924	0	1
3-6-12	0.02235	477.0	2379	762	285	1	0.048404694	477	5726	2283	0	1
3-6-13	0.00632	488.588	584	90	138	1	0.039502382	488.588	5063	1805	0	1
3-6-14	0.03789	478.224	4522	1424	557	1	0.215825796	478.224	26899	10978	0	1
3-6-15	1.72466	539.796	152823	55983	13733	1	9.384262323	539.796	730951	309509	0	1
3-6-16	0.66742	550.466	63348	17791	11950	1	6.840775967	550.466	545242	225880	0	1
3-6-17	2.12598	560.218	173350	38794	44987	1	38.11250257	560.218	2581531	1030307	0	1
3-6-18	0.00375	477.918	377	109	38	1	0.01087904	477.918	1299	528	0	1
3-6-19	9.28452	570.582	749311	268797	64177	1	101.725889	570.582	7154283	2960123	0	1
3-6-20	0.09239	498.646	9872	3269	1174	1	0.286472321	498.646	30371	12546	0	1
3-6-21	0.00166	412.388	159	39	17	1	0.002705574	412.388	317	108	0	1
3-6-22	0.04973	477.306	5677	1944	673	1	0.204900742	477.306	23472	9865	0	1
3-6-23	0.63268	518.762	60977	20385	7953	1	4.438118458	518.762	366805	152851	0	1
3-6-24	0.01385	478.836	1782	594	175	1	0.026962996	478.836	3837	1536	0	1
3-6-25	0.06690	519.68	8421	2438	1497	1	6.594615221	519.68	591741	241921	0	1
3-6-26	0.00877	462.372	908	188	196	1	0.071053743	462.372	8939	3560	0	1
3-6-27	0.03573	503.828	4078	1305	481	1	0.163213968	503.828	18846	7888	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-6-28	0.00504	514.804	498	108	98	1	0.020756721	514.804	2922	984	0	1
3-6-29	0.00126	448.05	115	36	4	1	0.001601934	448.05	175	56	0	1
3-6-30	0.05682	463.596	6912	2350	839	1	0.147643328	463.596	16979	7045	0	1
3-6-31	0.04724	528.514	5006	1879	330	1	0.070075035	528.514	7917	3327	0	1
3-6-32	0.02797	468.166	2964	881	474	1	0.061085224	468.166	7161	2907	0	1
3-6-33	0.01393	469.084	1645	595	117	1	0.020366669	469.084	2645	1080	0	1
3-6-34	0.04683	477.306	4902	1313	934	1	0.188058615	477.306	19795	8038	0	1
3-6-35	0.01989	477.0	2110	455	461	1	0.146522999	477	16201	6417	0	1
3-6-36	0.00140	448.356	134	25	23	1	0.00289011	448.356	368	128	0	1
3-6-37	0.55763	523.944	50507	15331	8057	1	3.039162159	523.944	253436	103428	0	1
3-6-38	0.20934	519.986	20633	6934	2287	1	0.879183054	519.986	80651	33170	0	1
3-6-39	0.01013	467.554	1226	467	69	1	0.047050238	467.554	6092	2556	0	1
3-6-40	0.23673	508.092	25578	8777	2664	1	1.891731262	508.092	170485	72382	0	1
4-4-1	0.03043	504.45	3690	1114	373	1	0.350199699	504.45	40477	15455	0	1
4-4-2	0.00622	463.912	714	114	152	1	0.065692902	463.912	9290	3032	0	1
4-4-3	0.02195	546.518	2465	768	133	1	0.115879536	546.518	14748	5279	0	1
4-4-4	0.01720	417.274	2140	588	301	1	0.041971207	417.274	6207	2320	0	1
4-4-5	0.01684	443.184	2073	491	331	1	0.334470987	443.184	43426	15753	0	1
4-4-6	0.97980	519.69	92543	27251	9201	1	3.984819651	519.69	347996	127282	0	1
4-4-7	0.00440	422.15	501	67	128	1	0.032285929	422.15	4464	1434	0	1
4-4-8	0.12862	529.748	15795	3625	2660	1	0.921516657	529.748	100185	34703	0	1
4-4-9	0.01582	484.64	1691	374	241	1	0.036687374	484.64	4485	1436	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-4-10	0.06554	524.26	7022	1828	631	1	0.192270994	524.26	21142	6827	0	1
4-4-11	0.10193	494.086	10798	2037	1816	1	1.269380569	494.086	121631	42202	0	1
4-4-12	0.00385	452.936	450	87	77	1	0.048380613	452.936	7402	2500	0	1
4-4-13	0.00218	381.306	265	74	20	1	0.005977392	381.306	879	296	0	1
4-4-14	0.06941	505.062	7673	1789	1058	1	0.380433559	505.062	43386	14729	0	1
4-4-15	0.00838	437.696	901	214	145	1	0.023751259	437.696	2869	1016	0	1
4-4-16	0.01415	494.392	1514	243	334	1	0.091340303	494.392	12418	4146	0	1
4-4-17	0.00333	412.398	366	82	52	1	0.006701231	412.398	870	285	0	1
4-4-18	0.02854	498.656	3605	1077	258	1	0.070345402	498.656	9965	3568	0	1
4-4-19	0.00886	462.994	1124	354	62	1	0.0376544	462.994	5407	1956	0	1
4-4-20	0.02829	505.062	3617	1236	155	1	0.174962997	505.062	22637	8519	0	1
4-4-21	0.01119	464.218	1307	379	151	1	0.036860228	464.218	4823	1824	0	1
4-4-22	0.00996	432.82	1067	223	173	1	0.040139198	432.82	5022	1696	0	1
4-4-23	0.02494	458.73	2914	682	336	1	0.060294151	458.73	8852	3088	0	1
4-4-24	0.04779	468.176	5399	1144	962	1	0.247839451	468.176	29573	9871	0	1
4-4-25	0.00924	463.3	1104	308	116	1	0.019649506	463.3	2688	954	0	1
4-4-26	0.00423	438.308	476	122	53	1	0.005703688	438.308	760	258	0	1
4-4-27	0.19432	524.566	21808	5601	3223	1	2.135877848	524.566	208378	73781	0	1
4-4-28	0.00535	458.424	591	159	50	1	0.009063721	458.424	1174	385	0	1
4-4-29	0.03524	519.69	3596	841	328	1	0.075650454	519.69	8833	2808	0	1
4-4-30	0.05428	499.268	6301	1645	732	1	1.285841227	499.268	136122	49803	0	1
4-4-31	0.00441	396.852	473	136	36	1	0.005996943	396.852	739	256	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-4-32	0.10412	519.384	12321	4213	659	1	0.548202753	519.384	62283	23948	0	1
4-4-33	0.17106	507.184	17826	5391	1405	1	0.498007059	507.184	53291	19097	0	1
4-4-34	0.00483	422.456	522	116	81	1	0.012345791	422.456	1618	537	0	1
4-4-35	0.37980	525.484	39411	11703	3113	1	1.764440775	525.484	170649	60776	0	1
4-4-36	0.00844	447.754	891	204	125	1	0.019011021	447.754	2312	768	0	1
4-4-37	0.00078	391.67	77	14	8	1	0.002624512	391.67	385	122	0	1
4-4-38	0.08794	539.194	11330	2934	1643	1	3.387676239	539.194	318635	115659	0	1
4-4-39	0.00978	457.812	1111	324	84	1	0.03877449	457.812	5146	1875	0	1
4-4-40	0.00272	433.126	263	67	19	1	0.002993584	433.126	324	95	0	1
4-5-1	6.88510	646.486	546781	184265	54043	1	127.2714097	646.486	9160509	3725038	0	1
4-5-2	0.03750	605.03	4040	974	879	1	0.276591539	605.03	30160	11540	0	1
4-5-3	0.00931	512.06	1134	315	147	1	0.087039471	512.06	11418	4212	0	1
4-5-4	0.07292	579.12	7968	2226	1334	1	2.468143702	579.12	209624	81842	0	1
4-5-5	0.00274	502.92	265	86	9	1	0.004204988	502.92	498	179	0	1
4-5-6	0.20354	542.234	19611	5383	3182	1	3.927212477	542.234	297836	117155	0	1
4-5-7	2.26691	660.196	202900	70267	14987	1	186.0240314	660.196	13352279	5259134	0	1
4-5-8	0.81629	604.418	70152	17470	13876	1	9.25275135	604.418	662930	255256	0	1
4-5-9	2.25328	687.024	190321	58140	29834	1	41.31071067	687.024	2718678	1113891	0	1
4-5-10	1.11646	694.042	97865	27184	14912	1	56.34259462	694.042	3813135	1403605	0	1
4-5-11	4.92033	649.832	415766	132813	49041	1	218.8327296	649.832	15181517	6117635	0	1
4-5-12	0.06229	631.552	6731	1647	1192	1	2.156793833	631.552	188591	70062	0	1
4-5-13	0.20933	641.61	15854	1601	3923	1	6.510981321	641.61	530531	195910	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-5-14	9.45623	692.206	719613	233178	82513	1	132.8496215	692.206	8928786	3588951	0	1
4-5-15	1.73668	609.294	154657	55904	9047	1	30.73901248	609.294	2232969	910199	0	1
4-5-16	2.42554	669.948	211818	50788	51861	1	720.2884314	669.948	46432897	18111311	0	1
4-5-17	3.30482	620.27	256895	78592	32588	1	14.72490144	620.27	1040782	400858	0	1
4-5-18	0.02383	671.784	2061	238	536	1	0.352339268	671.784	39489	13126	0	1
4-5-19	7.90952	660.196	598867	226449	26136	1	65.83228588	660.196	4103819	1673162	0	1
4-5-20	0.02676	589.178	3031	758	451	1	1.61418891	589.178	154914	58533	0	1
4-5-21	0.03576	563.574	3660	848	722	1	0.530525923	563.574	54829	20936	0	1
4-5-22	5.25951	680.618	435800	128030	68235	1	80.87640762	680.618	5660821	2252179	0	1
4-5-23	0.39871	623.922	37865	11064	5207	1	5.774795771	623.922	438356	172235	0	1
4-5-24	11.63965	707.752	855225	317378	49011	1	338.1650429	707.752	22586645	9154232	0	1
4-5-25	2.62495	749.208	186029	39367	32740	1	21.61422229	749.208	1339338	450437	0	1
4-5-26	0.03277	532.482	4219	1415	427	1	0.172165632	532.482	22145	9023	0	1
4-5-27	0.13597	605.03	14166	3186	2522	1	5.947342873	605.03	473333	176758	0	1
4-5-28	1.94011	649.22	154166	55836	10116	1	16.95570683	649.22	1168818	466413	0	1
4-5-29	27.59201	674.518	2115582	705360	253123	1	>1000s (timeout)	-	46323411	0	0	0
4-5-30	0.10140	552.598	10956	2434	2681	1	7.610235453	552.598	577380	226022	0	1
4-5-31	3.00853	661.42	226014	81924	9234	1	14.03875971	661.42	911839	352530	0	1
4-5-32	0.18541	619.964	18288	4720	3297	1	3.114572287	619.964	259515	99578	0	1
4-5-33	4.28619	711.71	368736	113373	52580	1	587.592494	711.71	44277589	18299341	0	1
4-5-34	0.05978	544.376	7854	2790	564	1	0.194181204	544.376	25626	10405	0	1
4-5-35	0.25027	589.178	23329	5906	4090	1	1.61938405	589.178	134629	51796	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-5-36	4.90291	670.866	384175	128047	36888	1	52.70074201	670.866	3465675	1401304	0	1
4-5-37	1.22006	640.08	104318	39402	5090	1	34.14835286	640.08	2394507	981554	0	1
4-5-38	0.03968	595.278	4499	1192	782	1	2.544050932	595.278	219413	84292	0	1
4-5-39	0.17910	624.534	16633	3769	3273	1	8.220197201	624.534	582402	226565	0	1
4-5-40	0.58857	665.99	55778	11808	12985	1	107.3589256	665.99	7617223	2876864	0	1
5-4-1	0.89858	681.536	82983	23017	11739	1	4.08723712	681.536	316796	116531	0	1
5-4-2	0.02973	565.104	2785	437	563	1	0.587619066	565.104	61024	21211	0	1
5-4-3	0.06030	564.492	5825	1152	1229	1	0.473529816	564.492	46874	16508	0	1
5-4-4	2.92240	692.512	215632	46299	36913	1	37.88721418	692.512	2411870	864532	0	1
5-4-5	18.72755	760.49	1374438	422582	120844	1	166.3321679	760.49	10904196	4073901	0	1
5-4-6	2.52852	704.712	220935	61175	29675	1	42.86473441	704.712	3043632	1131060	0	1
5-4-7	0.01975	615.394	2255	731	139	1	0.993292809	615.394	107404	41239	0	1
5-4-8	0.64407	714.158	61197	20986	3096	1	14.7350862	714.158	1091054	423528	0	1
5-4-9	3.33315	830.896	247548	60652	35825	1	21.70005369	830.896	1355508	465261	0	1
5-4-10	0.00401	529.442	403	101	45	1	0.009104729	529.442	1109	362	0	1
5-4-11	5.57003	728.48	451266	135025	48438	1	205.4702287	728.48	13717423	5059581	0	1
5-4-12	0.52830	821.45	42957	13507	1101	1	0.909316301	821.45	73951	24572	0	1
5-4-13	0.18259	703.488	18244	3872	3572	1	114.794749	703.488	7894339	2901000	0	1
5-4-14	10.32187	739.762	863565	265220	92644	1	251.9697776	739.762	18675437	7052286	0	1
5-4-15	0.36467	631.552	36456	10590	3374	1	1.911403656	631.552	182583	65940	0	1
5-4-16	0.00762	611.742	763	154	120	1	0.066623211	611.742	8128	2618	0	1
5-4-17	0.48993	749.82	43841	11452	3588	1	1.396947384	749.82	117816	38967	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
5-4-18	0.00507	488.904	476	73	74	1	0.016077518	488.904	1859	531	0	1
5-4-19	63.35968	831.202	4472986	1551097	95844	1	177.5114563	831.202	11117793	4037645	0	1
5-4-20	0.04337	631.246	3726	535	679	1	0.565487385	631.246	55588	18646	0	1
5-4-21	6.22721	733.968	420192	99179	43336	1	29.56291366	733.968	1941970	623034	0	1
5-4-22	1.52408	662.644	130623	30511	19275	1	69.14099789	662.644	4958134	1789209	0	1
5-4-23	0.38303	682.454	39737	13767	1895	1	7.919127226	682.454	665798	257374	0	1
5-4-24	0.04783	713.852	4536	1137	562	1	0.119051456	713.852	11761	3894	0	1
5-4-25	0.01087	534.318	1076	168	229	1	0.240949392	534.318	26274	8657	0	1
5-4-26	1.50916	749.514	121607	42177	3658	1	3.526032209	749.514	280164	102531	0	1
5-4-27	0.07160	642.222	6470	1409	828	1	0.262664795	642.222	26962	8649	0	1
5-4-28	1.33354	605.03	104745	15207	22306	1	51.6840961	605.03	3545078	1174015	0	1
5-4-29	0.19770	647.71	22688	6019	3001	1	8.082309723	647.71	701738	256516	0	1
5-4-30	1.56774	703.488	128741	17504	39347	1	151.551029	703.488	10162512	3697759	0	1
5-4-31	0.02279	637.346	1974	425	212	1	0.058953762	637.346	6271	1860	0	1
5-4-32	7.41029	723.91	543678	151839	51264	1	35.00478077	723.91	2185267	753387	0	1
5-4-33	1.77350	699.224	153974	51315	6038	1	8.398181438	699.224	643631	234142	0	1
5-4-34	0.02245	554.74	2212	627	162	1	0.066476345	554.74	7359	2434	0	1
5-4-35	0.33093	756.532	31291	6834	5341	1	5.121948004	756.532	391079	136704	0	1
5-4-36	3.49373	683.066	271686	42967	73996	1	69.85796189	683.066	4752584	1703873	0	1
5-4-37	0.21891	663.256	20793	2014	7298	1	26.54446888	663.256	1839627	655047	0	1
5-4-38	0.03866	627.288	4124	1070	505	1	0.392706871	627.288	41862	14580	0	1
5-4-39	4.09826	729.398	342055	117222	21303	1	101.7327824	729.398	6990938	2612486	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
5-4-40	7.99695	708.976	624636	162454	86974	1	67.26782179	708.976	4662624	1705266	0	1
3-7-1	0.01995	521.037	1886	562	254	1	0.053025723	521.037	5532	2177	0	1
3-7-2	0.00153	523.179	150	52	3	1	0.002647161	523.179	327	125	0	1
3-7-3	0.19085	590.239	16358	3271	4132	1	10.44716287	590.239	770369	318215	0	1
3-7-4	0.08734	598.767	9524	2352	2264	1	0.621997118	598.767	58572	23928	0	1
3-7-5	0.63534	609.131	54685	17536	8318	1	2.689071417	609.131	208729	87186	0	1
3-7-6	0.23900	649.669	20539	2485	7321	1	17.59469748	649.669	1238954	490717	0	1
3-7-7	0.02518	579.875	2408	405	711	1	0.612033129	579.875	61307	23301	0	1
3-7-8	0.10203	621.025	11617	3699	1790	1	0.270401478	621.025	28395	11576	0	1
3-7-9	0.16804	629.247	18876	5170	3216	1	23.12637925	629.247	1985711	834060	0	1
3-7-10	0.00876	517.385	964	320	123	1	0.011892319	517.385	1402	583	0	1
3-7-11	1.07381	614.925	89665	30934	13122	1	10.8029995	614.925	749631	327445	0	1
3-7-12	0.05730	665.827	5673	1445	1215	1	0.685179472	665.827	62649	25596	0	1
3-7-13	0.01225	569.511	1196	391	118	1	0.024624109	569.511	2934	1154	0	1
3-7-14	0.05517	646.629	6019	2206	502	1	0.143395901	646.629	14928	6166	0	1
3-7-15	6.85011	669.479	563050	189177	59467	1	164.3403156	669.479	11878429	5074895	0	1
3-7-16	0.05642	583.833	5477	1492	901	1	1.457620144	583.833	116180	46262	0	1
3-7-17	0.01365	568.593	1477	431	239	1	0.145766497	568.593	15213	6170	0	1
3-7-18	0.37366	599.685	33467	8431	8216	1	2.707816124	599.685	204880	84753	0	1
3-7-19	1.13429	638.693	100686	25110	26032	1	31.72030354	638.693	2269471	958816	0	1
3-7-20	3.20285	660.033	238655	76035	32928	1	54.05637026	660.033	3684151	1576914	0	1
3-7-21	0.00734	532.319	820	272	122	1	0.014176846	532.319	1604	679	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-7-22	2.15304	652.117	183718	64381	23620	1	49.80393004	652.117	3544796	1530493	0	1
3-7-23	0.05209	548.783	5778	2102	489	1	0.107486248	548.783	12976	5478	0	1
3-7-24	0.00911	563.717	995	364	91	1	0.019061327	563.717	2238	950	0	1
3-7-25	0.01699	573.775	1591	545	187	1	0.028956175	573.775	3033	1267	0	1
3-7-26	0.37554	615.537	32591	7922	7373	1	6.598883152	615.537	480910	202148	0	1
3-7-27	9.23514	649.057	708445	223850	133031	1	341.3253248	649.057	21619835	9363876	0	1
3-7-28	0.20827	604.561	21031	6710	3805	1	2.662564754	604.561	208417	88204	0	1
3-7-29	5.58027	634.429	452141	154767	68771	1	213.6512232	634.429	14102537	6162566	0	1
3-7-30	0.01578	557.923	1645	559	176	1	0.050721645	557.923	5474	2245	0	1
3-7-31	0.01755	614.925	1581	438	235	1	0.066245079	614.925	7615	2975	0	1
3-7-32	1.59162	630.165	129947	49091	11119	1	6.094625473	630.165	444677	192154	0	1
3-7-33	0.00650	584.445	587	151	84	1	0.04252553	584.445	4804	1835	0	1
3-7-34	0.01364	589.627	1226	387	135	1	0.028285265	589.627	2863	1087	0	1
3-7-35	0.06453	614.313	6024	1259	1225	1	3.76450491	614.313	294728	116202	0	1
3-7-36	3.57425	656.381	289950	110871	26017	1	17.99311757	656.381	1256795	549868	0	1
3-7-37	0.30772	598.767	28139	8661	4760	1	2.875068426	598.767	237252	100924	0	1
3-7-38	2.83346	639.305	222944	75395	37529	1	77.46420527	639.305	4924171	2134856	0	1
3-7-39	0.14053	625.289	14420	2621	4553	1	18.54264712	625.289	1382923	565413	0	1
3-7-40	2.13819	649.975	172682	54303	29680	1	32.09459281	649.975	2226543	960129	0	1
3-8-1	0.16663	688.82	13124	4531	1648	1	0.526412487	688.82	40538	17178	0	1
3-8-2	0.84400	727.542	62243	17312	15390	1	5.018782377	727.542	329078	140424	0	1
3-8-3	0.07001	712.302	6358	2433	459	1	0.113339663	712.302	10909	4667	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-8-4	2.79647	733.03	227831	78427	34511	1	90.03467822	733.03	5951476	2630924	0	1
3-8-5	0.93176	707.12	72983	19812	18075	1	7.53661108	707.12	475744	201727	0	1
3-8-6	14.09169	772.65	1023580	380346	135254	1	312.956681	772.65	19236494	8569994	0	1
3-8-7	0.01495	651.036	1215	421	117	1	0.022011757	651.036	1951	808	0	1
3-8-8	0.08700	681.21	8372	2566	1317	1	1.60194993	681.21	126688	53239	0	1
3-8-9	0.02961	686.698	2987	710	667	1	0.819857359	686.698	75018	30893	0	1
3-8-10	0.08684	681.21	7869	2791	934	1	0.238213539	681.21	21057	9008	0	1
3-8-11	0.35637	726.93	27434	8382	4903	1	1.819089651	726.93	133979	56659	0	1
3-8-12	0.65421	686.392	55612	19703	7149	1	14.12738109	686.392	918820	398997	0	1
3-8-13	1.08966	726.624	87294	25048	20992	1	8.660809278	726.624	562929	243881	0	1
3-8-14	0.08001	686.698	8838	3299	720	1	0.282245159	686.698	28048	12023	0	1
3-8-15	16.12467	809.23	1170717	381092	190946	1	938.4711123	809.23	60290460	25669413	0	1
3-8-16	0.03364	686.086	2843	985	372	1	0.093762636	686.086	8282	3598	0	1
3-8-17	0.06328	736.988	5828	1414	1355	1	1.571014881	736.988	125668	52093	0	1
3-8-18	0.03977	645.854	4101	1585	250	1	0.103116274	645.854	11437	4985	0	1
3-8-19	61.10510	782.402	3992531	1424306	532103	1	506.3936722	782.402	30683951	13556443	0	1
3-8-20	0.31479	690.962	25785	8722	3906	1	1.071637154	690.962	79830	34162	0	1
3-8-21	0.26424	716.872	22662	5043	5100	1	54.9788444	716.872	3554875	1478062	0	1
3-8-22	90.11945	799.784	6282862	2220011	976773	1	>1000s (timeout)	-	39721031	0	0	0
3-8-23	0.03203	656.524	3392	1168	446	1	0.174129486	656.524	17509	7437	0	1
3-8-24	0.45186	722.36	37242	13202	3690	1	18.75991631	722.36	1226925	535554	0	1
3-8-25	197.08044	813.8	13085864	4070889	2793292	1	>1000s (timeout)	-	37402807	0	0	0

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-8-26	5.14114	737.906	359626	114886	75107	1	43.463269	737.906	2622699	1150742	0	1
3-8-27	5.53884	742.17	386108	116173	80776	1	69.62078166	742.17	4184363	1815711	0	1
3-8-28	9.85063	716.26	681249	257422	82062	1	53.30819559	716.26	3206475	1438192	0	1
3-8-29	42.15343	753.146	2839649	919961	514808	1	>1000s (timeout)	-	38880729	0	0	0
3-8-30	6.98657	722.36	495637	188506	51216	1	188.5561404	722.36	12114450	5381258	0	1
3-8-31	0.09471	671.458	8044	1701	2035	1	3.359853029	671.458	256816	104139	0	1
3-8-32	35.01865	809.842	2563114	871161	380034	1	>1000s (timeout)	-	42142117	0	0	0
3-8-33	0.68309	759.858	53009	17272	7780	1	10.97035289	759.858	722955	314485	0	1
3-8-34	3.37360	747.046	251234	71449	50722	1	646.6626871	747.046	41580696	18158322	0	1
3-8-35	0.12726	717.79	10425	2546	1798	1	1.204687119	717.79	95630	38952	0	1
3-8-36	0.40084	721.748	34720	9545	6434	1	13.32785702	721.748	961980	401833	0	1
3-8-37	0.63985	696.756	51128	17795	7056	1	4.573154926	696.756	297167	129491	0	1
3-8-38	22.38134	746.74	1545071	572439	171350	1	199.299109	746.74	12582754	5518596	0	1
3-8-39	0.46085	721.748	37698	14491	3319	1	2.61088109	721.748	186849	83098	0	1
3-8-40	0.41310	695.532	35054	12834	3661	1	1.718738317	695.532	127901	56020	0	1
4-6-1	7.11918	848.258	544268	113971	169579	1	>1000s (timeout)	-	39482948	0	0	0
4-6-2	5.79602	765.652	423799	145270	48252	1	484.642647	765.652	28604660	12182090	0	1
4-6-3	3.10251	739.436	202048	68107	22099	1	9.356283665	739.436	590337	241443	0	1
4-6-4	2.10782	801.008	149162	20104	44443	1	>1000s (timeout)	-	58537533	20346879	0	1
4-6-5	20.47899	734.254	1400988	312454	397053	1	>1000s (timeout)	-	39317733	0	0	0
4-6-6	0.07343	731.214	6261	1166	1667	1	1.446038961	731.214	121472	47926	0	1
4-6-7	3.71107	785.156	309848	122380	21046	1	>1000s (timeout)	-	45098544	0	0	0

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-6-8	0.41584	653.484	36369	8353	9051	1	22.18371892	653.484	1550284	647207	0	1
4-6-9	26.96692	782.116	1834229	399724	514533	1	>1000s (timeout)	-	38091417	0	0	0
4-6-10	79.55970	837.894	4881624	1105334	1227136	1	>1000s (timeout)	-	38290592	0	0	0
4-6-11	0.66460	755.594	50759	17534	4899	1	3.301627398	755.594	231394	92390	0	1
4-6-12	0.07828	647.69	7178	2087	930	1	0.974218369	647.69	81649	32557	0	1
4-6-13	0.14334	779.362	15487	5043	1880	1	18.74139237	779.362	1495390	609558	0	1
4-6-14	8.83128	781.81	676032	242397	73849	1	126.9700007	781.81	8715064	3738455	0	1
4-6-15	0.39813	704.998	32892	6772	9997	1	4.954282999	704.998	335719	134951	0	1
4-6-16	0.25106	699.204	22698	6580	4453	1	3.194336176	699.204	246351	102408	0	1
4-6-17	6.61469	735.172	483758	135174	96476	1	>1000s (timeout)	-	40724945	0	0	0
4-6-18	121.19033	828.448	7727528	2654635	836226	1	>1000s (timeout)	-	37736505	0	0	0
4-6-19	0.44513	724.196	38460	11636	7064	1	16.96586847	724.196	1142073	478854	0	1
4-6-20	26.24828	791.868	1930733	526831	468801	1	>1000s (timeout)	-	40876914	0	0	0
4-6-21	35.66756	827.836	2575592	895666	304056	1	>1000s (timeout)	-	41310035	0	0	0
4-6-22	4.57050	761.388	326353	96981	52841	1	112.8528569	761.388	6804050	2742141	0	1
4-6-23	5.71181	709.262	459219	150451	53265	1	98.13134217	709.262	7197093	3036012	0	1
4-6-24	5.45248	805.884	366819	69456	106699	1	452.7284017	805.884	27903028	11327918	0	1
4-6-25	0.19810	705.304	16196	2940	4155	1	5.812423706	705.304	414398	154832	0	1
4-6-26	0.84243	731.214	65231	16087	15491	1	10.98593903	731.214	708378	290007	0	1
4-6-27	27.02818	807.414	1856105	448975	399098	1	>1000s (timeout)	-	41496710	0	0	0
4-6-28	1.61315	745.536	126217	31804	31178	1	65.95584345	745.536	3908659	1616967	0	1
4-6-29	16.94593	745.23	1099646	269430	268280	1	656.5077639	745.23	37519936	15331329	0	1

表 B.1 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-6-30	543.63920	874.168	34296373	13097552	2765107	1	>1000s (timeout)	-	38073805	0	0	0
4-6-31	120.15875	854.052	7805245	2682156	964034	1	>1000s (timeout)	-	38130515	0	0	0
4-6-32	0.44288	752.554	39986	14438	2990	1	2.46531868	752.554	201170	83664	0	1
4-6-33	186.35746	790.338	11417021	3520044	1868852	1	>1000s (timeout)	-	37858550	0	0	0
4-6-34	27.76256	734.866	1903796	519492	414532	1	219.4361391	734.866	13471963	5531802	0	1
4-6-35	42.12669	822.348	3181038	801244	848838	1	>1000s (timeout)	-	39043018	0	0	0
4-6-36	9.42597	788.828	583272	161763	91315	1	148.2750075	788.828	8530496	3401994	0	1
4-6-37	253.78767	811.372	17293045	5610829	2600761	1	>1000s (timeout)	-	40421278	0	0	0
4-6-38	1.08237	838.812	79735	22915	13718	1	9.532859564	838.812	625594	247581	0	1
4-6-39	0.42621	755.9	43216	13166	6275	1	265.0821476	755.9	18820574	7784544	0	1
4-6-40	5.88840	796.744	413533	116068	75799	1	53.84481502	796.744	3174717	1267784	0	1

B.2 精确求解器 (双下界策略) 结果对比

表 B.2 精确求解器 (UBHMM + 双下界策略 + BFS) 实验结果对比

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-3-1	0.00081	242.473	78	14	7	1	0.00081	242.473	106	26	0	1
3-3-2	0.00043	211.687	37	7	2	1	0.00038	211.687	39	9	0	1
3-3-3	0.00020	190.653	14	3	0	1	0.00018	190.653	14	3	0	1
3-3-4	0.00047	226.621	42	8	2	1	0.00053	226.621	56	13	0	1
3-3-5	0.00123	262.589	126	20	15	1	0.00212	262.589	307	82	0	1
3-3-6	0.00075	226.315	72	15	3	1	0.00080	226.315	89	22	0	1
3-3-7	0.00102	242.167	110	26	7	1	0.00114	242.167	155	44	0	1
3-3-8	0.00151	262.283	163	37	11	2	0.00193	262.283	253	72	0	2
3-3-9	0.00061	211.381	66	17	3	1	0.00058	211.381	75	22	0	1
3-3-10	0.00023	191.265	15	1	1	1	0.00027	191.265	24	3	0	1
3-3-11	0.00111	226.315	114	23	8	1	0.00131	226.315	171	43	0	1
3-3-12	0.00050	211.075	44	5	5	1	0.00067	211.075	82	19	0	1
3-3-13	0.00077	236.985	70	6	14	1	0.00119	236.985	166	43	0	1
3-3-14	0.00094	241.555	98	22	8	1	0.00107	241.555	145	41	0	1
3-3-15	0.00028	170.231	25	6	0	1	0.00027	170.231	25	6	0	1
3-3-16	0.00066	200.711	57	11	3	2	0.00068	200.711	71	17	0	2
3-3-17	0.00076	226.315	79	13	11	1	0.00121	226.315	176	51	0	1
3-3-18	0.00041	216.869	32	7	1	1	0.00035	216.869	33	8	0	1
3-3-19	0.00028	190.959	20	4	1	1	0.00026	190.959	24	6	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-3-20	0.00075	237.597	75	16	7	1	0.00087	237.597	104	29	0	1
3-3-21	0.00068	205.893	65	12	5	1	0.00082	205.893	111	28	0	1
3-3-22	0.00062	221.439	55	12	3	1	0.00066	221.439	75	19	0	1
3-3-23	0.00020	175.719	13	3	0	1	0.00017	175.719	13	3	0	1
3-3-24	0.00059	226.315	54	6	6	2	0.00110	226.315	155	38	0	2
3-3-25	0.00050	221.439	42	7	5	1	0.00059	221.439	70	19	0	1
3-3-26	0.00023	175.719	15	3	1	1	0.00022	175.719	19	5	0	1
3-3-27	0.00031	180.595	21	4	1	1	0.00028	180.595	25	6	0	1
3-3-28	0.00074	237.291	74	17	3	1	0.00082	237.291	95	25	0	1
3-3-29	0.00054	222.051	47	6	5	1	0.00068	222.051	79	18	0	1
3-3-30	0.00117	257.713	113	20	10	1	0.00141	257.713	181	45	0	1
3-3-31	0.00045	195.835	38	5	5	1	0.00060	195.835	66	17	0	1
3-3-32	0.00043	201.323	37	8	1	1	0.00040	201.323	44	11	0	1
3-3-33	0.00006	129.387	1	0	0	1	0.00005	129.387	1	0	0	1
3-3-34	0.00034	169.925	26	6	1	1	0.00029	169.925	27	7	0	1
3-3-35	0.00081	206.199	93	20	7	1	0.00098	206.199	143	40	0	1
3-3-36	0.00015	160.173	8	1	0	1	0.00014	160.173	8	1	0	1
3-3-37	0.00048	200.711	45	9	3	1	0.00053	200.711	59	15	0	1
3-3-38	0.00063	221.745	61	9	8	2	0.00077	221.745	98	26	0	2
3-3-39	0.00018	149.809	11	2	0	1	0.00015	149.809	11	2	0	1
3-3-40	0.00107	262.283	96	19	6	1	0.00118	262.283	139	35	0	1
3-4-1	0.00086	253.592	88	21	9	1	0.00101	253.592	131	42	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-4-2	0.01542	371.554	2003	564	223	1	0.04394	371.554	6691	2421	0	1
3-4-3	0.00294	330.404	356	84	43	3	0.00807	330.404	1290	449	0	3
3-4-4	0.00477	309.676	620	176	76	1	0.01074	309.676	1733	653	0	1
3-4-5	0.00056	264.874	56	16	2	1	0.00058	264.874	63	20	0	1
3-4-6	0.00285	320.958	324	87	35	1	0.00357	320.958	500	175	0	1
3-4-7	0.00072	294.13	73	16	6	1	0.00143	294.13	181	52	0	1
3-4-8	0.00629	330.404	845	271	49	1	0.01925	330.404	3084	1143	0	1
3-4-9	0.00121	273.402	132	22	28	1	0.00303	273.402	445	148	0	1
3-4-10	0.00525	325.528	635	155	83	2	0.01158	325.528	1737	606	0	2
3-4-11	0.00137	325.834	160	43	11	1	0.00224	325.834	325	106	0	1
3-4-12	0.00044	268.832	38	9	1	1	0.00051	268.832	49	12	0	1
3-4-13	0.00159	299.312	171	25	39	1	0.00583	299.312	869	282	0	1
3-4-14	0.00148	294.742	171	52	11	2	0.00179	294.742	254	89	0	2
3-4-15	0.00096	284.072	91	11	17	1	0.00197	284.072	272	78	0	1
3-4-16	0.00049	258.774	42	8	4	1	0.00057	258.774	66	20	0	1
3-4-17	0.00227	304.8	290	76	36	1	0.00596	304.8	1007	352	0	1
3-4-18	0.00432	309.982	515	140	59	1	0.00831	309.982	1158	413	0	1
3-4-19	0.00056	248.41	54	13	5	1	0.00050	248.41	59	18	0	1
3-4-20	0.00026	228.6	18	4	0	1	0.00024	228.6	18	4	0	1
3-4-21	0.00853	330.404	1045	252	150	1	0.01743	330.404	2544	882	0	1
3-4-22	0.00488	340.462	602	127	76	1	0.02143	340.462	3636	1203	0	1
3-4-23	0.00153	294.742	169	27	29	1	0.00283	294.742	428	136	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-4-24	0.00289	299.312	323	70	43	2	0.00460	299.312	657	210	0	2
3-4-25	0.00329	284.378	403	103	53	1	0.00939	284.378	1456	536	0	1
3-4-26	0.00255	273.402	291	87	22	4	0.00335	273.402	416	148	0	4
3-4-27	0.01954	351.132	2678	732	358	1	0.10009	351.132	16318	6003	0	1
3-4-28	0.00208	274.32	249	50	40	1	0.00627	274.32	1002	336	0	1
3-4-29	0.00173	320.958	196	39	33	1	0.00342	320.958	484	150	0	1
3-4-30	0.01004	320.04	1295	306	219	3	0.02749	320.04	4378	1558	0	3
3-4-31	0.00042	253.592	45	10	4	1	0.00053	253.592	68	20	0	1
3-4-32	0.00149	268.832	179	49	19	1	0.00180	268.832	255	90	0	1
3-4-33	0.00224	304.8	255	54	40	1	0.00594	304.8	815	273	0	1
3-4-34	0.00126	263.956	134	36	14	1	0.00233	263.956	297	104	0	1
3-4-35	0.00336	289.56	440	121	47	1	0.00737	289.56	1138	406	0	1
3-4-36	0.00114	259.386	120	24	17	2	0.00185	259.386	253	81	0	2
3-4-37	0.00218	325.528	241	52	42	1	0.00535	325.528	749	259	0	1
3-4-38	0.01167	330.098	1501	457	113	3	0.02413	330.098	3606	1318	0	3
3-4-39	0.00247	294.436	327	73	59	1	0.00410	294.436	671	234	0	1
3-4-40	0.00218	315.776	236	54	30	1	0.00262	315.776	355	115	0	1
3-5-1	0.00414	375.053	482	107	83	1	0.01255	375.053	1850	675	0	1
3-5-2	0.00112	303.117	120	34	11	1	0.00119	303.117	151	51	0	1
3-5-3	0.08675	451.253	9927	3112	1047	1	0.18045	451.253	23534	9000	0	1
3-5-4	0.00715	432.055	817	228	124	1	0.01220	432.055	1607	602	0	1
3-5-5	0.13346	442.113	16133	4324	2868	1	0.53045	442.113	71284	27633	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-5-6	0.00356	411.327	395	85	69	1	0.00784	411.327	1051	360	0	1
3-5-7	0.00274	426.567	310	74	42	1	0.00979	426.567	1385	500	0	1
3-5-8	0.01257	411.021	1492	400	225	1	0.11067	411.021	16402	6468	0	1
3-5-9	0.00609	411.327	659	141	131	1	0.02730	411.327	3731	1310	0	1
3-5-10	0.00101	323.845	94	29	4	1	0.00091	323.845	108	36	0	1
3-5-11	0.00424	410.715	517	130	76	1	0.01567	410.715	2499	936	0	1
3-5-12	0.02150	384.499	2564	802	227	1	0.06550	384.499	9424	3703	0	1
3-5-13	0.00243	374.441	247	61	29	1	0.00431	374.441	536	177	0	1
3-5-14	0.03331	396.393	3695	471	1212	2	0.46022	396.393	61011	22250	0	2
3-5-15	0.01156	380.541	1401	426	172	2	0.02472	380.541	3483	1375	0	2
3-5-16	0.00789	384.805	958	254	156	1	0.02360	384.805	3278	1245	0	1
3-5-17	0.04078	415.897	4970	1542	682	2	0.31321	415.897	39598	16026	0	2
3-5-18	0.00122	359.507	135	44	7	1	0.00113	359.507	142	51	0	1
3-5-19	0.02398	401.269	2953	879	388	2	0.09707	401.269	13593	5350	0	2
3-5-20	0.00300	410.103	334	108	19	1	0.00348	410.103	455	163	0	1
3-5-21	0.00303	389.681	352	57	90	1	0.03858	389.681	6074	2187	0	1
3-5-22	0.01735	400.963	2010	664	170	1	0.02509	400.963	3303	1279	0	1
3-5-23	0.00786	406.451	911	224	114	1	0.03204	406.451	4781	1752	0	1
3-5-24	0.00490	339.391	554	154	73	1	0.01054	339.391	1492	562	0	1
3-5-25	0.05247	421.691	6363	1891	736	1	0.20999	421.691	32203	12573	0	1
3-5-26	0.02155	410.715	2190	382	509	1	0.08616	410.715	11274	4032	0	1
3-5-27	0.00324	380.235	352	119	20	1	0.00488	380.235	600	227	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-5-28	0.00597	375.053	691	212	86	2	0.00863	375.053	1089	425	0	2
3-5-29	0.00221	370.177	239	41	49	1	0.01278	370.177	1977	681	0	1
3-5-30	0.00516	385.723	606	205	50	1	0.00925	385.723	1261	507	0	1
3-5-31	0.37021	513.437	38975	10778	5830	1	2.05419	513.437	256990	98515	0	1
3-5-32	0.00067	328.721	59	12	6	1	0.00175	328.721	215	69	0	1
3-5-33	0.00085	333.903	77	23	4	1	0.00102	333.903	104	36	0	1
3-5-34	0.01016	400.351	1091	177	297	1	0.04419	400.351	5864	2152	0	1
3-5-35	0.00461	349.143	532	153	53	1	0.01073	349.143	1478	542	0	1
3-5-36	0.03581	440.889	4381	1410	473	1	0.09669	440.889	12976	5146	0	1
3-5-37	0.00579	411.021	684	166	120	1	0.03571	411.021	5150	1932	0	1
3-5-38	0.00779	369.565	872	254	132	1	0.01199	369.565	1503	579	0	1
3-5-39	0.12559	473.511	13527	4027	1649	1	0.47800	473.511	62507	23947	0	1
3-5-40	0.04478	405.839	5549	1572	873	1	0.46684	405.839	67179	27139	0	1
3-6-1	1.13346	549.242	105275	37208	11960	1	10.19327	549.242	1103269	470007	0	1
3-6-2	0.01252	463.596	1273	283	250	1	0.05932	463.596	7491	2883	0	1
3-6-3	0.00065	438.298	58	17	3	1	0.00063	438.298	61	20	0	1
3-6-4	0.09885	519.986	11221	3043	2082	1	3.48422	519.986	395729	162644	0	1
3-6-5	0.10718	545.896	10704	2556	2288	1	0.49232	545.896	56805	21946	0	1
3-6-6	0.02814	503.828	3208	838	675	1	0.37835	503.828	48011	19765	0	1
3-6-7	0.17736	498.952	18548	5876	2972	1	1.03150	498.952	124064	51905	0	1
3-6-8	0.25037	514.192	26226	8708	3434	1	0.96608	514.192	116664	48882	0	1
3-6-9	0.14073	581.558	15191	4651	2321	2	0.46823	581.558	57026	22736	0	2

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-6-10	0.00209	462.678	241	71	29	1	0.01000	462.678	1434	560	0	1
3-6-11	0.06343	519.374	7003	2576	541	1	0.14543	519.374	18027	7625	0	1
3-6-12	0.02190	477.0	2318	748	269	2	0.04903	477	5574	2226	0	2
3-6-13	0.00659	488.588	586	91	137	1	0.03807	488.588	4927	1753	0	1
3-6-14	0.03480	478.224	4101	1313	468	1	0.16932	478.224	22651	9239	0	1
3-6-15	1.60220	539.796	145012	53069	13170	1	6.47428	539.796	684399	290085	0	1
3-6-16	0.60891	550.466	61849	17435	11559	1	4.26956	550.466	515703	213527	0	1
3-6-17	1.99641	560.218	171801	38503	44622	2	26.11169	560.218	2525976	1008849	0	2
3-6-18	0.00357	477.918	346	102	33	1	0.01037	477.918	1251	511	0	1
3-6-19	8.57939	570.582	706511	254111	59600	1	64.28930	570.582	6561492	2716778	0	1
3-6-20	0.08675	498.646	9380	3109	1110	1	0.23315	498.646	28291	11684	0	1
3-6-21	0.00163	412.388	156	40	16	1	0.00269	412.388	312	108	0	1
3-6-22	0.05032	477.306	5614	1932	651	1	0.18796	477.306	23287	9796	0	1
3-6-23	0.58820	518.762	59124	19925	7449	2	3.11993	518.762	351844	146666	0	2
3-6-24	0.01363	478.836	1718	576	165	1	0.02505	478.836	3683	1477	0	1
3-6-25	0.06759	519.68	8287	2424	1437	1	4.48861	519.68	569229	233216	0	1
3-6-26	0.00963	462.372	978	199	229	1	0.06238	462.372	7847	3124	0	1
3-6-27	0.03694	503.828	4094	1307	473	2	0.14703	503.828	18687	7810	0	2
3-6-28	0.00484	514.804	465	99	91	1	0.02052	514.804	2878	968	0	1
3-6-29	0.00130	448.05	115	36	4	1	0.00165	448.05	175	56	0	1
3-6-30	0.05556	463.596	6564	2245	773	1	0.12571	463.596	16039	6654	0	1
3-6-31	0.04739	528.514	4922	1844	324	2	0.06986	528.514	7888	3310	0	2

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-6-32	0.02839	468.166	2957	880	471	1	0.06082	468.166	7121	2891	0	1
3-6-33	0.01348	469.084	1539	565	95	2	0.01806	469.084	2380	972	0	2
3-6-34	0.04809	477.306	4904	1315	937	3	0.17064	477.306	19612	7979	0	3
3-6-35	0.02072	477.0	2133	456	467	2	0.13148	477	16399	6484	0	2
3-6-36	0.00149	448.356	133	28	20	1	0.00268	448.356	329	116	0	1
3-6-37	0.53780	523.944	49158	15063	7654	1	2.19844	523.944	240879	98342	0	1
3-6-38	0.20421	519.986	19681	6656	2135	1	0.64284	519.986	75515	31173	0	1
3-6-39	0.01094	467.554	1271	480	67	2	0.04423	467.554	6106	2555	0	2
3-6-40	0.23868	508.092	25414	8716	2690	2	1.36607	508.092	164877	70002	0	2
4-4-1	0.03006	504.45	3528	1064	357	1	0.28360	504.45	37839	14450	0	1
4-4-2	0.00649	463.912	708	114	152	1	0.06328	463.912	8657	2836	0	1
4-4-3	0.02249	546.518	2425	758	128	1	0.11851	546.518	14565	5220	0	1
4-4-4	0.01640	417.274	2025	577	247	2	0.04078	417.274	5854	2190	0	2
4-4-5	0.01697	443.184	2031	487	312	2	0.28860	443.184	42855	15537	0	2
4-4-6	0.89823	519.69	86714	25824	8339	1	2.86581	519.69	322756	118417	0	1
4-4-7	0.00452	422.15	491	64	127	1	0.03212	422.15	4149	1330	0	1
4-4-8	0.12894	529.748	15287	3521	2517	2	0.69031	529.748	93162	32200	0	2
4-4-9	0.01596	484.64	1663	368	242	1	0.03685	484.64	4366	1406	0	1
4-4-10	0.06768	524.26	6810	1788	597	1	0.17285	524.26	20514	6637	0	1
4-4-11	0.10350	494.086	10629	2033	1737	2	0.98700	494.086	117440	40708	0	2
4-4-12	0.00431	452.936	483	97	72	4	0.04869	452.936	7325	2477	0	4
4-4-13	0.00223	381.306	262	70	20	3	0.00557	381.306	802	266	0	3

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-4-14	0.07259	505.062	7733	1793	1088	1	0.33229	505.062	42427	14427	0	1
4-4-15	0.00792	437.696	829	194	136	1	0.02045	437.696	2444	864	0	1
4-4-16	0.01440	494.392	1494	243	327	1	0.08877	494.392	11777	3939	0	1
4-4-17	0.00328	412.398	350	79	47	1	0.00609	412.398	762	247	0	1
4-4-18	0.02953	498.656	3584	1072	251	3	0.07119	498.656	9957	3564	0	3
4-4-19	0.00877	462.994	1094	342	63	2	0.03785	462.994	5295	1917	0	2
4-4-20	0.02736	505.062	3393	1167	133	1	0.15168	505.062	21631	8156	0	1
4-4-21	0.01116	464.218	1253	366	139	2	0.03646	464.218	4597	1739	0	2
4-4-22	0.01017	432.82	1068	230	163	2	0.04047	432.82	4880	1651	0	2
4-4-23	0.02478	458.73	2797	654	312	1	0.05751	458.73	8561	2979	0	1
4-4-24	0.04791	468.176	5231	1128	900	2	0.21787	468.176	27292	9125	0	2
4-4-25	0.00927	463.3	1076	304	104	2	0.02013	463.3	2597	919	0	2
4-4-26	0.00458	438.308	506	128	50	2	0.00619	438.308	793	263	0	2
4-4-27	0.19259	524.566	20705	5310	3040	2	1.56784	524.566	192722	68180	0	2
4-4-28	0.00545	458.424	584	155	54	2	0.00877	458.424	1118	366	0	2
4-4-29	0.03429	519.69	3331	784	288	1	0.07334	519.69	8428	2681	0	1
4-4-30	0.05553	499.268	6226	1625	707	3	1.05060	499.268	128214	46672	0	3
4-4-31	0.00432	396.852	449	131	29	1	0.00590	396.852	718	248	0	1
4-4-32	0.10160	519.384	11520	3943	624	1	0.46079	519.384	58333	22465	0	1
4-4-33	0.17304	507.184	17752	5404	1337	1	0.44567	507.184	53237	19084	0	1
4-4-34	0.00463	422.456	497	112	74	1	0.01174	422.456	1564	519	0	1
4-4-35	0.37164	525.484	38250	11449	2898	2	1.31185	525.484	162662	58017	0	2

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-4-36	0.00860	447.754	887	201	128	1	0.01948	447.754	2290	760	0	1
4-4-37	0.00078	391.67	76	14	8	1	0.00248	391.67	346	108	0	1
4-4-38	0.08921	539.194	11296	2963	1572	1	2.41874	539.194	318576	115639	0	1
4-4-39	0.00943	457.812	1048	305	79	1	0.03757	457.812	4882	1782	0	1
4-4-40	0.00281	433.126	255	65	17	2	0.00294	433.126	314	91	0	2
4-5-1	6.17776	646.486	527762	179178	51099	4	88.62308	646.486	8835675	3599054	0	4
4-5-2	0.03908	605.03	3978	964	855	1	0.28375	605.03	29699	11349	0	1
4-5-3	0.00947	512.06	1122	315	138	1	0.08358	512.06	10918	4029	0	1
4-5-4	0.07159	579.12	7598	2092	1331	1	1.91906	579.12	204361	79780	0	1
4-5-5	0.00247	502.92	236	77	10	1	0.00351	502.92	409	150	0	1
4-5-6	0.19570	542.234	18169	5069	2833	1	2.52067	542.234	268297	105630	0	1
4-5-7	2.25058	660.196	195311	67730	14086	1	134.02980	660.196	13197852	5198205	0	1
4-5-8	0.82618	604.418	69287	17240	13714	1	7.05218	604.418	651496	250783	0	1
4-5-9	2.15043	687.024	183930	55951	29302	1	27.01964	687.024	2595526	1062945	0	1
4-5-10	1.08876	694.042	95539	26779	14268	1	35.80731	694.042	3558780	1315176	0	1
4-5-11	4.71356	649.832	401371	128327	47588	1	143.11589	649.832	14607105	5886866	0	1
4-5-12	0.06303	631.552	6485	1572	1173	1	1.61817	631.552	176647	65719	0	1
4-5-13	0.22182	641.61	16022	1604	4025	1	4.28927	641.61	490124	179938	0	1
4-5-14	8.50374	692.206	687886	223356	77993	1	86.63628	692.206	8346772	3355211	0	1
4-5-15	1.70650	609.294	153919	55818	8712	2	19.00428	609.294	2070420	843305	0	2
4-5-16	2.43434	669.948	211035	50313	52122	3	521.51244	669.948	46318305	18063693	0	3
4-5-17	3.32355	620.27	252725	77582	31511	3	11.22770	620.27	1027753	395756	0	3

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-5-18	0.02429	671.784	2004	232	505	1	0.30982	671.784	39317	13060	0	1
4-5-19	7.60223	660.196	580233	219658	25123	1	43.39188	660.196	3941898	1607248	0	1
4-5-20	0.02697	589.178	2989	749	442	1	1.33142	589.178	153815	58191	0	1
4-5-21	0.03556	563.574	3710	876	718	1	0.43786	563.574	52841	20222	0	1
4-5-22	5.03123	680.618	427841	126309	66208	1	54.96343	680.618	5356837	2130756	0	1
4-5-23	0.38597	623.922	37807	11033	5231	1	4.08722	623.922	436098	171359	0	1
4-5-24	10.39962	707.752	830618	308102	47943	1	236.84753	707.752	21836246	8846388	0	1
4-5-25	2.62940	749.208	185920	39045	33112	1	15.49850	749.208	1330115	447404	0	1
4-5-26	0.03230	532.482	4052	1378	383	1	0.14835	532.482	21064	8597	0	1
4-5-27	0.13266	605.03	13655	3032	2516	1	4.26635	605.03	462151	172249	0	1
4-5-28	1.83648	649.22	153936	55694	10249	1	12.61913	649.22	1167155	465806	0	1
4-5-29	24.61535	674.518	2119652	708288	251142	2	987.63157	674.518	95633189	39002737	0	2
4-5-30	0.10554	552.598	11201	2465	2778	2	5.59978	552.598	572984	224193	0	2
4-5-31	2.91679	661.42	216249	78617	8588	1	10.20016	661.42	882909	341487	0	1
4-5-32	0.19279	619.964	17984	4637	3244	1	2.31632	619.964	253028	97110	0	1
4-5-33	3.95883	711.71	360774	111895	49901	2	411.03106	711.71	43623818	18030927	0	2
4-5-34	0.05394	544.376	6914	2487	442	2	0.15780	544.376	22869	9288	0	2
4-5-35	0.24716	589.178	22425	5763	3769	1	1.21391	589.178	130220	50084	0	1
4-5-36	4.85197	670.866	383491	127838	36937	1	34.34194	670.866	3388444	1369210	0	1
4-5-37	1.13083	640.08	102268	38573	5054	1	24.00106	640.08	2357155	965823	0	1
4-5-38	0.03909	595.278	4373	1148	775	1	1.70308	595.278	204003	78315	0	1
4-5-39	0.15847	624.534	15670	3579	3092	2	5.46314	624.534	555960	216244	0	2

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-5-40	0.59380	665.99	55087	11554	12921	1	71.08975	665.99	7485923	2827540	0	1
5-4-1	0.91798	681.536	83381	23291	11509	4	3.12208	681.536	316508	116408	0	4
5-4-2	0.02969	565.104	2724	442	532	1	0.48681	565.104	59288	20655	0	1
5-4-3	0.06210	564.492	5861	1132	1267	3	0.42597	564.492	45847	16138	0	3
5-4-4	2.92965	692.512	215429	46884	36021	1	26.22305	692.512	2338315	838334	0	1
5-4-5	17.00171	760.49	1369664	424732	114163	1	115.12181	760.49	10622268	3968295	0	1
5-4-6	2.43837	704.712	210048	58993	27020	3	28.80112	704.712	2872600	1069200	0	3
5-4-7	0.01928	615.394	2212	722	129	1	0.84421	615.394	106952	41063	0	1
5-4-8	0.63958	714.158	58958	20104	3175	1	10.66768	714.158	1034139	401190	0	1
5-4-9	3.34123	830.896	243081	60272	34153	2	16.40253	830.896	1325750	455482	0	2
5-4-10	0.00385	529.442	386	99	40	2	0.00850	529.442	1026	339	0	2
5-4-11	5.24348	728.48	444284	132449	48606	2	148.42980	728.48	13656671	5036922	0	2
5-4-12	0.52889	821.45	42303	13335	1117	1	0.77972	821.45	72007	23986	0	1
5-4-13	0.18326	703.488	17484	3697	3399	2	87.04337	703.488	7699527	2824207	0	2
5-4-14	10.02680	739.762	856849	263497	91418	1	193.17489	739.762	18520110	6994889	0	1
5-4-15	0.37095	631.552	35878	10428	3317	1	1.55866	631.552	179613	64826	0	1
5-4-16	0.00777	611.742	757	150	122	1	0.06930	611.742	8117	2614	0	1
5-4-17	0.49975	749.82	43130	11160	3643	3	1.09627	749.82	115630	38183	0	3
5-4-18	0.00500	488.904	464	86	56	1	0.01285	488.904	1552	443	0	1
5-4-19	62.14277	831.202	4463967	1550608	91770	1	136.63746	831.202	11101248	4031745	0	1
5-4-20	0.04406	631.246	3747	533	683	2	0.51213	631.246	55491	18609	0	2
5-4-21	5.75824	733.968	408983	97722	41187	1	21.88756	733.968	1833805	590315	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
5-4-22	1.52380	662.644	129983	30444	19370	2	44.06469	662.644	4463411	1607915	0	2
5-4-23	0.37547	682.454	38413	13443	1651	2	5.69461	682.454	644500	249353	0	2
5-4-24	0.04845	713.852	4489	1127	557	1	0.11580	713.852	11564	3839	0	1
5-4-25	0.01057	534.318	1022	156	221	1	0.21617	534.318	24597	8103	0	1
5-4-26	1.45216	749.514	118528	40983	3681	1	2.82263	749.514	272162	99420	0	1
5-4-27	0.07120	642.222	6267	1359	798	1	0.23750	642.222	25463	8108	0	1
5-4-28	1.31227	605.03	104971	15495	22092	2	33.79875	605.03	3527645	1168189	0	2
5-4-29	0.19508	647.71	21601	5699	2849	2	5.58161	647.71	658325	240419	0	2
5-4-30	1.55409	703.488	127360	17492	38623	2	103.64410	703.488	9839229	3580658	0	2
5-4-31	0.02489	637.346	2104	452	224	1	0.05896	637.346	6269	1860	0	1
5-4-32	7.09808	723.91	543317	152089	50556	1	24.36550	723.91	2175905	750679	0	1
5-4-33	1.74475	699.224	152264	50753	6085	1	6.22851	699.224	633784	230776	0	1
5-4-34	0.02294	554.74	2218	633	156	1	0.06512	554.74	7125	2363	0	1
5-4-35	0.33302	756.532	30790	6711	5246	1	3.52015	756.532	367764	128308	0	1
5-4-36	3.38299	683.066	265711	42675	71114	2	46.49894	683.066	4421911	1585279	0	2
5-4-37	0.22455	663.256	20674	2097	7110	1	18.22013	663.256	1773975	632206	0	1
5-4-38	0.03784	627.288	3963	1042	460	1	0.34008	627.288	39518	13722	0	1
5-4-39	3.96589	729.398	332246	113124	21574	1	70.07566	729.398	6586234	2455887	0	1
5-4-40	8.21938	708.976	615301	160598	84934	2	49.26759	708.976	4598133	1681989	0	2
3-7-1	0.02054	521.037	1906	568	252	2	0.05147	521.037	5494	2161	0	2
3-7-2	0.00115	523.179	101	32	3	1	0.00229	523.179	278	105	0	1
3-7-3	0.19265	590.239	16153	3226	4112	1	7.09233	590.239	767160	317193	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-7-4	0.08718	598.767	9242	2294	2177	1	0.50260	598.767	55928	22818	0	1
3-7-5	0.62944	609.131	55137	17652	8474	3	1.95977	609.131	207421	86602	0	3
3-7-6	0.24218	649.669	20421	2458	7383	2	12.55546	649.669	1238504	490577	0	2
3-7-7	0.02550	579.875	2371	410	676	1	0.49843	579.875	61268	23301	0	1
3-7-8	0.09691	621.025	10682	3524	1390	1	0.20939	621.025	26675	10897	0	1
3-7-9	0.17012	629.247	18356	5018	3101	4	15.00859	629.247	1902520	798466	0	4
3-7-10	0.00837	517.385	910	303	115	1	0.01087	517.385	1295	538	0	1
3-7-11	1.03366	614.925	88497	30622	12862	1	7.78589	614.925	740821	323696	0	1
3-7-12	0.05875	665.827	5691	1450	1219	1	0.53571	665.827	62593	25571	0	1
3-7-13	0.01183	569.511	1135	376	103	1	0.02170	569.511	2599	1019	0	1
3-7-14	0.05536	646.629	5895	2190	429	1	0.12132	646.629	14624	6035	0	1
3-7-15	6.85761	669.479	570787	192150	60137	1	107.83126	669.479	11877272	5074519	0	1
3-7-16	0.05529	583.833	5222	1419	856	1	1.04926	583.833	109789	43788	0	1
3-7-17	0.01193	568.593	1224	360	185	1	0.11954	568.593	14147	5755	0	1
3-7-18	0.37404	599.685	32235	8369	7438	1	2.06895	599.685	191681	79343	0	1
3-7-19	1.11990	638.693	98174	24416	25520	1	22.33397	638.693	2214718	935547	0	1
3-7-20	3.01230	660.033	237098	75924	32094	1	33.07077	660.033	3591126	1538354	0	1
3-7-21	0.00714	532.319	731	240	109	1	0.01313	532.319	1478	624	0	1
3-7-22	2.09295	652.117	174752	61436	22076	1	31.16665	652.117	3214016	1388412	0	1
3-7-23	0.04733	548.783	5224	1889	465	1	0.09463	548.783	11826	4998	0	1
3-7-24	0.00826	563.717	866	329	61	1	0.01553	563.717	1892	810	0	1
3-7-25	0.01628	573.775	1598	553	179	1	0.02798	573.775	3030	1267	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-7-26	0.38250	615.537	32489	7765	7632	1	4.62317	615.537	475965	200011	0	1
3-7-27	9.15826	649.057	714113	225316	134552	1	221.22093	649.057	21619713	9363822	0	1
3-7-28	0.20700	604.561	20704	6646	3660	2	2.02839	604.561	202001	85600	0	2
3-7-29	5.46269	634.429	449849	153111	70051	1	135.44715	634.429	14031016	6131203	0	1
3-7-30	0.01618	557.923	1633	555	175	1	0.04315	557.923	4824	1975	0	1
3-7-31	0.01760	614.925	1571	441	223	1	0.06251	614.925	7324	2857	0	1
3-7-32	1.54514	630.165	126844	48084	10543	1	3.93381	630.165	426411	184303	0	1
3-7-33	0.00647	584.445	585	151	84	1	0.04116	584.445	4443	1695	0	1
3-7-34	0.01279	589.627	1114	357	110	1	0.02290	589.627	2412	922	0	1
3-7-35	0.06515	614.313	5960	1233	1227	1	2.57904	614.313	292254	115418	0	1
3-7-36	3.35325	656.381	273124	104708	24201	1	11.34356	656.381	1165451	510188	0	1
3-7-37	0.29200	598.767	28014	8752	4539	1	2.11415	598.767	229061	97526	0	1
3-7-38	2.80549	639.305	218966	73774	37256	2	54.68362	639.305	4816486	2087179	0	2
3-7-39	0.14601	625.289	14127	2644	4362	2	11.58496	625.289	1198984	490684	0	2
3-7-40	2.07156	649.975	171177	53644	29687	1	21.59966	649.975	2187573	943238	0	1
3-8-1	0.16786	688.82	13152	4550	1643	2	0.45168	688.82	40537	17181	0	2
3-8-2	0.76700	727.542	59629	16610	14751	1	3.59210	727.542	307900	131424	0	1
3-8-3	0.06385	712.302	5777	2249	362	2	0.09301	712.302	9169	3937	0	2
3-8-4	2.76277	733.03	226844	77798	34837	1	57.90422	733.03	5699328	2519396	0	1
3-8-5	0.92196	707.12	72631	19886	17659	1	5.27669	707.12	474894	201436	0	1
3-8-6	12.62793	772.65	985210	364617	132550	2	205.35208	772.65	18072126	8050446	0	2
3-8-7	0.01337	651.036	1083	381	103	1	0.01748	651.036	1521	634	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-8-8	0.08789	681.21	8351	2571	1305	2	1.19881	681.21	124859	52626	0	2
3-8-9	0.02949	686.698	2881	681	643	1	0.57918	686.698	65989	27164	0	1
3-8-10	0.08949	681.21	8038	2863	953	2	0.20924	681.21	20383	8723	0	2
3-8-11	0.34989	726.93	27316	8369	4870	2	1.40315	726.93	132312	55997	0	2
3-8-12	0.57429	686.392	50908	18051	6506	3	8.87953	686.392	817763	354933	0	3
3-8-13	1.10949	726.624	88024	25451	20836	5	6.49651	726.624	561898	243439	0	5
3-8-14	0.08063	686.698	8787	3290	699	1	0.23586	686.698	27947	11987	0	1
3-8-15	14.59244	809.23	1131104	370541	180705	1	589.84753	809.23	57510164	24497926	0	1
3-8-16	0.03404	686.086	2737	960	339	1	0.08602	686.086	7687	3349	0	1
3-8-17	0.06229	736.988	5503	1298	1327	1	1.20812	736.988	122861	50995	0	1
3-8-18	0.03590	645.854	3653	1437	200	1	0.08534	645.854	9722	4259	0	1
3-8-19	56.62841	782.402	3964780	1416562	523298	3	347.87439	782.402	30167007	13322865	0	3
3-8-20	0.30600	690.962	25568	8561	3953	2	0.93281	690.962	81645	34619	0	2
3-8-21	0.27875	716.872	23687	5237	5435	1	37.41634	716.872	3539498	1472584	0	1
3-8-22	81.07487	799.784	6013614	2133600	918297	1	1000.00000	-	68213593	22476741	0	1
3-8-23	0.02810	656.524	2960	1060	327	1	0.14239	656.524	15535	6610	0	1
3-8-24	0.40979	722.36	35767	12725	3490	1	13.57920	722.36	1201750	524944	0	1
3-8-25	182.70514	813.8	12779993	4002480	2683708	2	1000.00000	-	36752923	0	0	0
3-8-26	4.95066	737.906	359157	114705	75115	1	32.33079	737.906	2615814	1147843	0	1
3-8-27	5.39969	742.17	372124	112833	76046	1	48.80297	742.17	3936490	1707183	0	1
3-8-28	9.21847	716.26	641387	242926	76103	2	35.06702	716.26	2966841	1330268	0	2
3-8-29	37.52696	753.146	2730595	886054	490010	1	1000.00000	-	84289508	35398466	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-8-30	6.43294	722.36	482007	183661	49576	1	126.79132	722.36	11588703	5152024	0	1
3-8-31	0.09580	671.458	7925	1714	1945	2	2.32062	671.458	235686	95785	0	2
3-8-32	32.93612	809.842	2513277	851970	375223	1	1000.00000	-	79722441	27395196	0	1
3-8-33	0.63726	759.858	49450	16121	7213	1	7.12933	759.858	668819	291157	0	1
3-8-34	3.31409	747.046	250309	70650	51209	1	422.44301	747.046	40729635	17786260	0	1
3-8-35	0.13169	717.79	10755	2731	1758	1	0.90141	717.79	86797	35305	0	1
3-8-36	0.36768	721.748	31865	8913	5632	2	8.05034	721.748	855883	358135	0	2
3-8-37	0.61105	696.756	50363	17519	6975	1	3.33668	696.756	293958	128147	0	1
3-8-38	21.55552	746.74	1556086	578029	169287	2	133.84107	746.74	12447553	5459498	0	2
3-8-39	0.45129	721.748	36913	14205	3248	1	1.87085	721.748	180687	80378	0	1
3-8-40	0.40929	695.532	34706	12744	3582	2	1.32382	695.532	125020	54828	0	2
4-6-1	7.09600	848.258	533383	111236	166281	3	1000.00000	-	42542270	1424590	0	1
4-6-2	5.25213	765.652	409512	139698	47572	1	311.88854	765.652	27233358	11585290	0	1
4-6-3	2.81558	739.436	190271	64855	20006	2	6.47908	739.436	531529	217750	0	2
4-6-4	2.14774	801.008	149832	20134	44759	1	695.72318	801.008	62218694	24041086	0	1
4-6-5	19.25941	734.254	1399346	312812	395601	1	1000.00000	-	82682421	32533010	0	1
4-6-6	0.07494	731.214	6172	1160	1610	1	1.15815	731.214	117632	46428	0	1
4-6-7	3.59186	785.156	307437	121320	21120	1	1000.00000	-	94152327	38789244	0	1
4-6-8	0.41871	653.484	35416	8257	8626	1	16.03754	653.484	1490123	623309	0	1
4-6-9	26.82648	782.116	1835498	396190	520109	2	1000.00000	-	78467961	30588294	0	2
4-6-10	72.59406	837.894	4771177	1080319	1199053	2	1000.00000	-	39929193	470144	0	2
4-6-11	0.64172	755.594	47923	16474	4599	1	2.26824	755.594	213513	85008	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-6-12	0.07625	647.69	7115	2085	900	1	0.77940	647.69	79313	31678	0	1
4-6-13	0.14168	779.362	15365	4999	1863	2	13.60397	779.362	1493584	608789	0	2
4-6-14	7.50192	781.81	637457	228084	70494	1	78.83158	781.81	8059534	3455983	0	1
4-6-15	0.37381	704.998	32180	6626	9782	2	3.67601	704.998	325470	131144	0	2
4-6-16	0.24335	699.204	22556	6501	4499	1	2.44608	699.204	245672	102128	0	1
4-6-17	6.33032	735.172	471846	133051	92148	5	1000.00000	-	84550014	33453963	0	5
4-6-18	105.57790	828.448	7398689	2541062	804774	1	1000.00000	-	78576819	31920338	0	1
4-6-19	0.43281	724.196	37611	11527	6699	2	12.16259	724.196	1105351	464360	0	2
4-6-20	24.18991	791.868	1857905	514375	438784	1	1000.00000	-	74878289	25541983	0	1
4-6-21	32.79873	827.836	2579475	899993	299018	2	773.07728	827.836	72384347	30913664	0	2
4-6-22	4.32524	761.388	307813	92338	48479	1	71.52868	761.388	6053405	2439749	0	1
4-6-23	5.68127	709.262	460149	150478	53690	2	66.85268	709.262	7135414	3009491	0	2
4-6-24	5.57278	805.884	368392	69566	106602	2	305.05797	805.884	27902179	11327549	0	2
4-6-25	0.19911	705.304	15931	2867	4134	1	3.87531	705.304	401449	150942	0	1
4-6-26	0.83545	731.214	64175	15845	15184	2	7.69475	731.214	689644	282151	0	2
4-6-27	25.75729	807.414	1802897	439605	381580	2	922.15261	807.414	82265246	33036723	0	2
4-6-28	1.59336	745.536	124051	31443	30588	2	44.52325	745.536	3686594	1526663	0	2
4-6-29	15.25805	745.23	1065811	263520	256695	1	428.11374	745.23	34679066	14179894	0	1
4-6-30	469.33239	874.168	33788429	12902163	2725715	2	1000.00000	-	38832398	0	0	0
4-6-31	105.27481	854.052	7568358	2612472	920837	1	1000.00000	-	67467927	22513717	0	1
4-6-32	0.40228	752.554	35194	12725	2566	1	1.56747	752.554	168984	70292	0	1
4-6-33	168.14782	790.338	11234483	3486339	1798849	1	1000.00000	-	70435163	24785973	0	1

表 B.2 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-6-34	26.84474	734.866	1879311	515047	405325	1	156.48018	734.866	13070455	5366645	0	1
4-6-35	39.81093	822.348	3083628	771018	831076	1	1000.00000	-	39403112	0	0	0
4-6-36	8.88705	788.828	563020	156142	88480	2	103.09482	788.828	8082889	3221421	0	2
4-6-37	240.94975	811.372	17366369	5637402	2607648	2	1000.00000	-	62348595	14805095	0	2
4-6-38	1.07794	838.812	79905	23048	13607	2	7.15093	838.812	625378	247514	0	2
4-6-39	0.40718	755.9	41014	12544	5893	3	167.57329	755.9	17039311	7036412	0	3
4-6-40	5.73631	796.744	401783	114364	71232	1	37.13712	796.744	3015962	1205846	0	1

B.3 启发式求解器 (LB_3) 结果对比

表 B.3 启发式求解器 (UBHMM + LB_3 + BFS) 实验结果对比

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-3-1	0.00047	242.473	47	11	2	1	0.00050	242.473	49	13	0	1
3-3-2	0.00029	211.687	26	6	0	1	0.00030	211.687	26	6	0	1
3-3-3	0.00021	190.653	14	3	0	1	0.00018	190.653	14	3	0	1
3-3-4	0.00037	226.621	36	8	1	1	0.00041	226.621	41	10	0	1
3-3-5	0.00070	262.589	72	13	8	1	0.00111	262.589	141	40	0	1
3-3-6	0.00059	226.315	60	12	3	1	0.00067	226.315	74	18	0	1
3-3-7	0.00064	242.167	65	15	4	1	0.00064	242.167	73	20	0	1
3-3-8	0.00097	268.077	109	25	8	1	0.00131	268.077	162	46	0	1
3-3-9	0.00034	211.381	32	8	1	1	0.00032	211.381	33	9	0	1
3-3-10	0.00021	191.265	15	1	1	1	0.00024	191.265	19	2	0	1
3-3-11	0.00065	226.315	67	14	5	1	0.00073	226.315	82	22	0	1
3-3-12	0.00031	211.075	29	7	0	1	0.00031	211.075	29	7	0	1
3-3-13	0.00046	236.985	44	9	3	1	0.00052	236.985	59	15	0	1
3-3-14	0.00046	241.555	51	15	0	1	0.00048	241.555	51	15	0	1
3-3-15	0.00023	170.231	17	4	0	1	0.00020	170.231	17	4	0	1
3-3-16	0.00047	205.893	46	10	2	1	0.00053	205.893	57	15	0	1
3-3-17	0.00056	226.315	57	10	7	1	0.00068	226.315	87	24	0	1
3-3-18	0.00030	216.869	26	6	0	1	0.00029	216.869	26	6	0	1
3-3-19	0.00022	190.959	16	4	0	1	0.00019	190.959	16	4	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-3-20	0.00038	237.597	37	9	2	1	0.00041	237.597	43	12	0	1
3-3-21	0.00053	205.893	55	12	2	1	0.00056	205.893	65	16	0	1
3-3-22	0.00040	221.439	39	9	1	1	0.00040	221.439	40	10	0	1
3-3-23	0.00015	175.719	8	2	0	1	0.00012	175.719	8	2	0	1
3-3-24	0.00039	226.621	39	6	3	1	0.00077	226.621	100	25	0	1
3-3-25	0.00038	221.439	35	9	1	1	0.00036	221.439	36	10	0	1
3-3-26	0.00015	175.719	9	2	0	1	0.00015	175.719	9	2	0	1
3-3-27	0.00028	180.595	21	4	1	1	0.00027	180.595	25	6	0	1
3-3-28	0.00053	237.291	56	15	0	1	0.00053	237.291	56	15	0	1
3-3-29	0.00028	222.051	22	4	1	1	0.00026	222.051	26	6	0	1
3-3-30	0.00056	257.713	62	15	2	1	0.00055	257.713	64	17	0	1
3-3-31	0.00029	195.835	24	6	0	1	0.00026	195.835	24	6	0	1
3-3-32	0.00030	201.323	26	5	1	1	0.00033	201.323	30	7	0	1
3-3-33	0.00006	129.387	1	0	0	1	0.00005	129.387	1	0	0	1
3-3-34	0.00024	169.925	20	5	0	1	0.00022	169.925	20	5	0	1
3-3-35	0.00055	206.199	63	16	2	1	0.00050	206.199	65	18	0	1
3-3-36	0.00015	160.173	8	1	0	1	0.00013	160.173	8	1	0	1
3-3-37	0.00034	200.711	30	7	0	1	0.00030	200.711	30	7	0	1
3-3-38	0.00048	222.051	49	8	7	1	0.00065	222.051	79	22	0	1
3-3-39	0.00017	149.809	11	2	0	1	0.00016	149.809	11	2	0	1
3-3-40	0.00058	262.283	62	14	3	1	0.00070	262.283	80	21	0	1
3-4-1	0.00053	253.592	60	17	3	1	0.00050	253.592	63	20	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-4-2	0.00607	371.554	764	218	67	1	0.01057	371.554	1483	510	0	1
3-4-3	0.00142	335.586	169	41	19	1	0.00288	335.586	433	150	0	1
3-4-4	0.00300	309.676	389	119	33	1	0.00569	309.676	876	329	0	1
3-4-5	0.00027	264.874	23	6	0	1	0.00025	264.874	23	6	0	1
3-4-6	0.00148	320.958	162	50	7	1	0.00148	320.958	189	65	0	1
3-4-7	0.00047	294.13	49	16	0	1	0.00044	294.13	49	16	0	1
3-4-8	0.00165	330.404	197	56	18	1	0.00365	330.404	543	198	0	1
3-4-9	0.00102	273.402	108	20	20	1	0.00211	273.402	288	96	0	1
3-4-10	0.00273	325.834	335	103	26	1	0.00392	325.834	550	202	0	1
3-4-11	0.00054	325.834	56	15	3	1	0.00068	325.834	87	28	0	1
3-4-12	0.00042	268.832	38	9	1	1	0.00046	268.832	43	10	0	1
3-4-13	0.00115	299.312	122	28	16	1	0.00238	299.312	329	108	0	1
3-4-14	0.00054	299.924	60	20	0	1	0.00052	299.924	60	20	0	1
3-4-15	0.00058	284.072	58	15	3	1	0.00062	284.072	76	22	0	1
3-4-16	0.00029	258.774	26	8	0	1	0.00026	258.774	26	8	0	1
3-4-17	0.00137	304.8	150	37	21	1	0.00235	304.8	347	120	0	1
3-4-18	0.00162	309.982	166	52	8	1	0.00158	309.982	205	72	0	1
3-4-19	0.00034	248.41	30	8	0	1	0.00030	248.41	30	8	0	1
3-4-20	0.00023	228.6	18	4	0	1	0.00021	228.6	18	4	0	1
3-4-21	0.00359	330.404	433	129	29	1	0.00508	330.404	691	244	0	1
3-4-22	0.00178	340.462	208	46	22	1	0.00370	340.462	576	190	0	1
3-4-23	0.00068	294.742	71	17	5	1	0.00088	294.742	110	34	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-4-24	0.00123	299.618	140	41	7	1	0.00144	299.618	179	60	0	1
3-4-25	0.00173	284.378	206	59	17	1	0.00237	284.378	338	120	0	1
3-4-26	0.00139	283.766	156	45	16	1	0.00157	283.766	195	70	0	1
3-4-27	0.00611	351.132	786	207	109	1	0.01645	351.132	2417	886	0	1
3-4-28	0.00096	274.32	109	31	3	1	0.00109	274.32	141	44	0	1
3-4-29	0.00070	320.958	81	25	3	1	0.00071	320.958	94	32	0	1
3-4-30	0.00455	320.652	575	141	88	1	0.00743	320.652	1101	387	0	1
3-4-31	0.00029	253.592	25	5	1	1	0.00034	253.592	31	8	0	1
3-4-32	0.00097	268.832	115	34	8	1	0.00096	268.832	128	44	0	1
3-4-33	0.00157	304.8	170	39	23	1	0.00287	304.8	371	123	0	1
3-4-34	0.00047	263.956	47	15	2	1	0.00050	263.956	59	21	0	1
3-4-35	0.00137	289.56	177	55	11	1	0.00220	289.56	330	118	0	1
3-4-36	0.00073	264.262	80	20	6	1	0.00092	264.262	119	38	0	1
3-4-37	0.00133	325.528	145	37	17	1	0.00213	325.528	271	92	0	1
3-4-38	0.00354	340.462	455	145	28	1	0.00674	340.462	992	366	0	1
3-4-39	0.00129	294.436	165	53	5	1	0.00128	294.436	186	64	0	1
3-4-40	0.00097	315.776	104	33	4	1	0.00094	315.776	113	39	0	1
3-5-1	0.00265	375.053	298	80	32	1	0.00495	375.053	671	242	0	1
3-5-2	0.00043	303.117	42	12	0	1	0.00041	303.117	42	12	0	1
3-5-3	0.02376	451.253	2684	867	219	1	0.03480	451.253	4478	1708	0	1
3-5-4	0.00258	432.055	287	84	42	1	0.00313	432.055	371	144	0	1
3-5-5	0.03949	442.113	4738	1521	500	1	0.08144	442.113	10927	4323	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-5-6	0.00233	411.327	249	55	41	1	0.00361	411.327	455	153	0	1
3-5-7	0.00129	426.567	143	42	12	1	0.00170	426.567	202	72	0	1
3-5-8	0.00616	411.021	695	165	113	1	0.02319	411.021	3359	1295	0	1
3-5-9	0.00341	411.327	374	104	45	1	0.00872	411.327	1156	392	0	1
3-5-10	0.00047	323.845	46	15	0	1	0.00043	323.845	46	15	0	1
3-5-11	0.00209	410.715	231	69	20	1	0.00321	410.715	432	161	0	1
3-5-12	0.00730	384.499	900	318	37	1	0.01417	384.499	1987	781	0	1
3-5-13	0.00103	374.441	103	33	2	1	0.00109	374.441	121	41	0	1
3-5-14	0.02581	400.045	2941	490	825	1	0.22433	400.045	27538	10047	0	1
3-5-15	0.00676	385.111	813	263	69	1	0.01054	385.111	1415	551	0	1
3-5-16	0.00409	384.805	500	150	60	1	0.00703	384.805	897	341	0	1
3-5-17	0.02321	416.203	2788	832	432	1	0.11118	416.203	15051	6022	0	1
3-5-18	0.00062	359.507	68	24	0	1	0.00059	359.507	68	24	0	1
3-5-19	0.00799	406.145	1007	331	95	1	0.02110	406.145	2864	1135	0	1
3-5-20	0.00144	410.103	149	48	3	1	0.00146	410.103	175	60	0	1
3-5-21	0.00163	389.681	182	42	30	1	0.01007	389.681	1484	534	0	1
3-5-22	0.00439	400.963	502	178	18	1	0.00475	400.963	597	228	0	1
3-5-23	0.00143	406.451	149	38	13	1	0.00327	406.451	451	159	0	1
3-5-24	0.00274	339.391	318	100	25	1	0.00425	339.391	563	209	0	1
3-5-25	0.00808	421.691	1011	346	59	1	0.01334	421.691	1910	735	0	1
3-5-26	0.00684	410.715	741	187	92	1	0.01443	410.715	1877	677	0	1
3-5-27	0.00168	380.235	188	68	7	1	0.00223	380.235	286	111	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-5-28	0.00221	375.665	243	85	13	1	0.00223	375.665	270	104	0	1
3-5-29	0.00155	370.177	163	39	18	1	0.00360	370.177	511	183	0	1
3-5-30	0.00290	385.723	347	119	27	1	0.00512	385.723	690	277	0	1
3-5-31	0.08308	513.437	9730	3057	958	1	0.27395	513.437	31070	11837	0	1
3-5-32	0.00063	328.721	57	13	5	1	0.00119	328.721	140	45	0	1
3-5-33	0.00048	333.903	44	15	0	1	0.00045	333.903	44	15	0	1
3-5-34	0.00734	400.351	799	165	169	1	0.02031	400.351	2596	944	0	1
3-5-35	0.00234	349.143	277	88	17	1	0.00500	349.143	701	258	0	1
3-5-36	0.00967	440.889	1178	395	95	1	0.01553	440.889	2062	809	0	1
3-5-37	0.00327	411.021	357	75	70	1	0.01257	411.021	1687	614	0	1
3-5-38	0.00501	369.565	577	195	53	1	0.00734	369.565	912	359	0	1
3-5-39	0.03181	473.511	3620	1220	282	1	0.06994	473.511	9186	3571	0	1
3-5-40	0.01921	405.839	2396	632	438	1	0.09834	405.839	13848	5515	0	1
3-6-1	0.33968	549.242	35104	13045	2994	1	2.95462	549.242	264274	113120	0	1
3-6-2	0.00797	463.596	833	202	140	1	0.02705	463.596	3419	1314	0	1
3-6-3	0.00047	438.298	38	12	0	1	0.00044	438.298	38	12	0	1
3-6-4	0.05036	519.986	5661	1727	761	1	0.93157	519.986	95445	39264	0	1
3-6-5	0.03269	545.896	3234	939	421	1	0.07313	545.896	8513	3233	0	1
3-6-6	0.01743	503.828	1926	548	315	1	0.09553	503.828	12354	5078	0	1
3-6-7	0.04680	498.952	5215	1685	780	1	0.16895	498.952	18162	7555	0	1
3-6-8	0.04507	514.192	5224	1676	832	1	0.11125	514.192	13228	5529	0	1
3-6-9	0.00616	585.516	646	243	39	1	0.00703	585.516	835	346	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-6-10	0.00137	462.678	148	47	9	1	0.00400	462.678	506	196	0	1
3-6-11	0.00835	519.374	897	314	114	1	0.00994	519.374	1119	476	0	1
3-6-12	0.00616	477.306	627	181	97	1	0.00977	477.306	1069	420	0	1
3-6-13	0.00331	488.588	298	72	34	1	0.00891	488.588	1102	389	0	1
3-6-14	0.00611	478.224	665	204	76	1	0.01352	478.224	1627	660	0	1
3-6-15	0.51882	539.796	50529	19059	3874	1	1.73226	539.796	159113	67994	0	1
3-6-16	0.06514	550.466	7132	2377	884	1	0.15948	550.466	16824	6964	0	1
3-6-17	0.82481	560.83	75639	21312	13727	1	7.37609	560.83	567609	228266	0	1
3-6-18	0.00267	477.918	244	64	28	1	0.00625	477.918	698	280	0	1
3-6-19	2.08817	570.582	175959	64083	14735	1	11.53329	570.582	890903	373711	0	1
3-6-20	0.01890	498.646	2089	732	199	1	0.03463	498.646	4179	1724	0	1
3-6-21	0.00103	412.388	97	33	3	1	0.00098	412.388	100	36	0	1
3-6-22	0.00975	477.306	1097	406	83	1	0.02072	477.306	2479	1038	0	1
3-6-23	0.20344	519.068	21432	7763	1912	1	0.93751	519.068	91503	38330	0	1
3-6-24	0.00313	478.836	380	137	22	1	0.00344	478.836	460	183	0	1
3-6-25	0.02238	519.68	2635	845	304	1	0.92062	519.68	94315	38489	0	1
3-6-26	0.00317	462.372	318	102	30	1	0.00461	462.372	530	204	0	1
3-6-27	0.02571	509.316	2870	970	258	1	0.07236	509.316	9209	3831	0	1
3-6-28	0.00135	514.804	130	42	6	1	0.00288	514.804	420	140	0	1
3-6-29	0.00066	448.05	66	24	0	1	0.00063	448.05	66	24	0	1
3-6-30	0.03108	463.596	3664	1299	346	1	0.05129	463.596	6592	2722	0	1
3-6-31	0.01429	529.126	1508	573	125	1	0.01941	529.126	2177	939	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-6-32	0.01030	468.166	1081	386	93	1	0.01648	468.166	1783	737	0	1
3-6-33	0.00288	473.96	327	134	2	1	0.00264	473.96	338	140	0	1
3-6-34	0.02020	478.224	1960	622	258	1	0.03561	478.224	3964	1615	0	1
3-6-35	0.01457	482.488	1427	304	293	1	0.06143	482.488	7448	2910	0	1
3-6-36	0.00093	448.356	86	20	8	1	0.00125	448.356	161	56	0	1
3-6-37	0.10105	523.944	10883	4016	828	1	0.29248	523.944	30077	12691	0	1
3-6-38	0.02715	519.986	2926	1120	171	1	0.04584	519.986	5373	2261	0	1
3-6-39	0.00181	473.348	191	71	5	1	0.00325	473.348	397	164	0	1
3-6-40	0.08362	508.704	9679	3474	903	1	0.34173	508.704	37667	15979	0	1
4-4-1	0.01049	504.45	1234	343	174	1	0.05811	504.45	7610	2824	0	1
4-4-2	0.00381	463.912	393	62	79	1	0.01687	463.912	2085	658	0	1
4-4-3	0.01411	546.518	1535	465	97	1	0.07082	546.518	9010	3290	0	1
4-4-4	0.00730	420.926	914	297	64	1	0.01349	420.926	1899	721	0	1
4-4-5	0.00844	448.366	993	215	181	1	0.08497	448.366	11979	4249	0	1
4-4-6	0.27501	519.69	29424	9802	1658	1	0.79308	519.69	81952	30651	0	1
4-4-7	0.00316	422.15	336	66	59	1	0.00956	422.15	1190	381	0	1
4-4-8	0.04693	530.054	5366	1422	526	1	0.11671	530.054	15494	5278	0	1
4-4-9	0.00668	484.64	669	149	91	1	0.01175	484.64	1313	419	0	1
4-4-10	0.02008	524.26	1967	509	135	1	0.03309	524.26	3625	1115	0	1
4-4-11	0.04334	498.656	4815	1358	466	1	0.20912	498.656	24645	9139	0	1
4-4-12	0.00221	457.812	234	51	31	1	0.01323	457.812	1860	640	0	1
4-4-13	0.00117	400.504	127	33	9	1	0.00310	400.504	406	130	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-4-14	0.02231	505.062	2454	725	167	1	0.04314	505.062	5526	1931	0	1
4-4-15	0.00413	437.696	408	92	64	1	0.00583	437.696	659	224	0	1
4-4-16	0.00741	494.392	785	156	134	1	0.02225	494.392	2778	901	0	1
4-4-17	0.00148	412.398	146	41	8	1	0.00145	412.398	154	49	0	1
4-4-18	0.00829	508.714	998	308	49	1	0.01597	508.714	2183	785	0	1
4-4-19	0.00332	463.912	374	109	22	1	0.01397	463.912	1859	645	0	1
4-4-20	0.00710	505.062	855	265	64	1	0.02492	505.062	3492	1304	0	1
4-4-21	0.00610	468.788	665	190	76	1	0.01465	468.788	1826	676	0	1
4-4-22	0.00357	437.696	374	105	34	1	0.00726	437.696	879	312	0	1
4-4-23	0.00399	458.73	450	134	26	1	0.00531	458.73	729	260	0	1
4-4-24	0.01420	468.482	1561	430	158	1	0.04119	468.482	5090	1757	0	1
4-4-25	0.00262	463.606	292	90	25	1	0.00446	463.606	572	213	0	1
4-4-26	0.00152	443.184	161	51	8	1	0.00162	443.184	198	70	0	1
4-4-27	0.07517	533.706	8655	2282	1131	1	0.48081	533.706	52835	18559	0	1
4-4-28	0.00261	463.606	278	78	21	1	0.00383	463.606	468	156	0	1
4-4-29	0.01260	519.69	1202	340	58	1	0.01782	519.69	1916	605	0	1
4-4-30	0.03845	509.632	4234	1079	503	1	0.53498	509.632	61100	22050	0	1
4-4-31	0.00188	396.852	193	62	7	1	0.00205	396.852	236	83	0	1
4-4-32	0.03444	519.384	3923	1213	394	1	0.12575	519.384	16441	6281	0	1
4-4-33	0.05302	507.184	5933	1797	544	1	0.12829	507.184	16083	5838	0	1
4-4-34	0.00203	422.456	208	56	17	1	0.00303	422.456	372	127	0	1
4-4-35	0.04775	534.318	5397	1834	256	1	0.15119	534.318	16962	6306	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-4-36	0.00392	447.754	392	97	46	1	0.00582	447.754	677	224	0	1
4-4-37	0.00080	391.67	76	15	7	1	0.00182	391.67	247	74	0	1
4-4-38	0.02226	539.194	2533	614	333	1	0.20733	539.194	26282	9442	0	1
4-4-39	0.00538	457.812	591	175	44	1	0.01781	457.812	2356	879	0	1
4-4-40	0.00125	438.002	118	40	1	1	0.00115	438.002	119	41	0	1
4-5-1	1.26858	655.014	113372	41068	7725	1	12.69762	655.014	1031349	420830	0	1
4-5-2	0.02879	605.03	3002	762	589	1	0.15652	605.03	15996	6108	0	1
4-5-3	0.00417	512.06	399	92	52	1	0.00976	512.06	1124	390	0	1
4-5-4	0.02684	579.12	2559	493	677	1	0.20349	579.12	19828	7330	0	1
4-5-5	0.00107	502.92	100	36	0	1	0.00098	502.92	100	36	0	1
4-5-6	0.05618	542.234	5515	1563	789	1	0.37468	542.234	37452	14816	0	1
4-5-7	0.78613	660.196	72585	24662	5977	1	44.25761	660.196	3216397	1280277	0	1
4-5-8	0.38159	604.418	33203	9163	5162	1	3.03388	604.418	233451	88954	0	1
4-5-9	0.51971	687.024	51037	11919	13478	1	5.23803	687.024	404550	160718	0	1
4-5-10	0.16387	694.042	15486	4464	1986	1	2.77196	694.042	248925	89890	0	1
4-5-11	0.84520	649.832	79020	26344	7584	1	18.45797	649.832	1403039	558802	0	1
4-5-12	0.04215	631.552	4267	1086	680	1	0.72240	631.552	68062	25708	0	1
4-5-13	0.04465	641.61	3923	682	703	1	0.68763	641.61	75006	28006	0	1
4-5-14	1.35142	692.206	115000	41419	7667	1	6.81301	692.206	539967	219802	0	1
4-5-15	0.08111	609.906	8121	2309	1281	1	0.63096	609.906	58335	22213	0	1
4-5-16	1.22131	675.742	110704	19651	35388	1	185.67174	675.742	11841072	4545840	0	1
4-5-17	0.49369	630.022	45426	14892	4815	1	1.39976	630.022	117352	46240	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-5-18	0.00325	671.784	322	95	28	1	0.00396	671.784	473	171	0	1
4-5-19	1.94961	660.196	160355	59036	10267	1	11.72850	660.196	839637	340603	0	1
4-5-20	0.01643	589.178	1715	405	271	1	0.42089	589.178	44057	16294	0	1
4-5-21	0.01537	563.574	1621	494	163	1	0.09011	563.574	10610	4089	0	1
4-5-22	0.40733	680.618	38422	14343	1877	1	5.00645	680.618	400992	164821	0	1
4-5-23	0.12809	623.922	13094	3273	2372	1	1.11136	623.922	102178	39256	0	1
4-5-24	2.22556	707.752	179049	63551	13144	1	41.36207	707.752	2834216	1125922	0	1
4-5-25	0.76501	749.208	56830	15730	5486	1	2.63361	749.208	197213	67194	0	1
4-5-26	0.00774	532.482	973	352	59	1	0.02882	532.482	3972	1626	0	1
4-5-27	0.04111	605.03	4159	911	745	1	0.71824	605.03	73729	27209	0	1
4-5-28	0.40657	649.22	36800	12970	2416	1	1.91819	649.22	157638	60788	0	1
4-5-29	9.54618	674.824	750415	241699	97531	1	241.20718	674.824	16528017	6644277	0	1
4-5-30	0.05870	553.21	5802	1323	1297	1	1.20339	553.21	105979	40970	0	1
4-5-31	0.58169	661.42	49080	17128	3380	1	2.58669	661.42	198911	77480	0	1
4-5-32	0.04773	619.964	4865	1403	687	1	0.28779	619.964	31013	12167	0	1
4-5-33	1.36424	712.016	127572	44347	11632	1	125.47507	712.016	9671111	3988838	0	1
4-5-34	0.01685	548.334	2165	803	98	1	0.03029	548.334	4269	1718	0	1
4-5-35	0.07434	589.178	7506	2361	769	1	0.22855	589.178	23908	9415	0	1
4-5-36	0.57329	670.866	50770	17764	3921	1	2.49915	670.866	208653	84084	0	1
4-5-37	0.43208	640.08	41017	14212	3770	1	10.01926	640.08	756744	307413	0	1
4-5-38	0.01525	595.278	1713	517	219	1	0.34351	595.278	37166	14535	0	1
4-5-39	0.05032	624.84	4751	944	1133	1	0.46694	624.84	44836	16675	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-5-40	0.16638	665.99	16068	4318	2394	1	5.08406	665.99	433604	163756	0	1
5-4-1	0.28427	692.206	26764	7863	2391	1	0.66616	692.206	64309	22686	0	1
5-4-2	0.01421	565.104	1357	316	175	1	0.07242	565.104	8123	2781	0	1
5-4-3	0.01832	569.98	1700	450	205	1	0.03272	569.98	3405	1203	0	1
5-4-4	0.63974	692.512	48511	13206	4213	1	2.58090	692.512	193371	66172	0	1
5-4-5	1.40340	760.49	115350	38214	6597	1	5.44089	760.49	410862	153142	0	1
5-4-6	0.44058	709.588	43736	13850	3227	1	3.06988	709.588	264396	97908	0	1
5-4-7	0.00912	615.394	1069	361	55	1	0.32411	615.394	36703	14257	0	1
5-4-8	0.30115	714.158	29756	9732	1902	1	4.17272	714.158	346880	132284	0	1
5-4-9	1.60617	835.772	125942	34507	14998	1	7.33561	835.772	496997	173530	0	1
5-4-10	0.00154	529.748	143	40	7	1	0.00240	529.748	273	87	0	1
5-4-11	1.77713	733.662	150781	40531	21876	1	39.25443	733.662	2691556	982921	0	1
5-4-12	0.10812	821.45	9376	2969	231	1	0.14409	821.45	13565	4449	0	1
5-4-13	0.12919	718.728	13400	2611	2777	1	60.84027	718.728	4431482	1603960	0	1
5-4-14	1.68428	739.762	140774	46389	10593	1	27.83973	739.762	2149369	829281	0	1
5-4-15	0.11645	631.552	12467	3717	1029	1	0.40525	631.552	43535	15434	0	1
5-4-16	0.00211	611.742	182	42	21	1	0.00330	611.742	333	107	0	1
5-4-17	0.14373	759.572	12790	3390	645	1	0.24247	759.572	22790	6984	0	1
5-4-18	0.00140	488.904	136	35	6	1	0.00191	488.904	222	65	0	1
5-4-19	23.56320	831.202	1634754	596623	20595	1	60.94570	831.202	3982649	1487315	0	1
5-4-20	0.02996	631.552	2500	354	420	1	0.22129	631.552	22335	7217	0	1
5-4-21	1.13486	733.968	87271	26328	4830	1	2.91214	733.968	224607	76579	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
5-4-22	0.47726	667.52	46132	12565	4923	1	10.52016	667.52	835978	303930	0	1
5-4-23	0.08581	682.76	10070	3829	145	1	1.82249	682.76	182828	71803	0	1
5-4-24	0.03249	713.852	3045	715	492	1	0.07570	713.852	7662	2620	0	1
5-4-25	0.00559	534.318	514	89	101	1	0.03020	534.318	3391	1094	0	1
5-4-26	0.21264	749.514	18784	6130	735	1	0.40851	749.514	37384	13147	0	1
5-4-27	0.00869	642.222	780	218	49	1	0.01326	642.222	1391	452	0	1
5-4-28	0.27987	610.212	25278	4647	4346	1	2.79319	610.212	237158	77758	0	1
5-4-29	0.05815	648.016	6154	1357	926	1	0.68689	648.016	66243	22386	0	1
5-4-30	0.68201	703.794	60997	12466	12868	1	18.19484	703.794	1333143	477128	0	1
5-4-31	0.00482	637.346	411	117	17	1	0.00521	637.346	499	157	0	1
5-4-32	1.01255	723.91	82577	25879	4071	1	2.67393	723.91	224652	78085	0	1
5-4-33	0.30084	699.224	30624	10597	1053	1	1.00157	699.224	94831	35455	0	1
5-4-34	0.00474	554.74	424	116	28	1	0.00878	554.74	894	290	0	1
5-4-35	0.05220	756.532	5073	1528	321	1	0.20504	756.532	20377	7011	0	1
5-4-36	1.15411	687.942	88632	18619	15700	1	6.85029	687.942	495317	169708	0	1
5-4-37	0.12986	663.256	12761	1846	3610	1	3.05067	663.256	243866	86095	0	1
5-4-38	0.01527	627.288	1536	393	171	1	0.08921	627.288	10207	3499	0	1
5-4-39	0.36156	729.398	32008	8051	4430	1	3.74467	729.398	270171	89258	0	1
5-4-40	3.00596	714.158	242386	69578	25056	1	15.90105	714.158	1172239	426691	0	1
3-7-1	0.00751	521.343	696	236	51	1	0.01278	521.343	1303	511	0	1
3-7-2	0.00102	523.179	96	35	0	1	0.00094	523.179	96	35	0	1
3-7-3	0.07287	590.239	6794	1986	1051	1	0.95865	590.239	90220	37332	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-7-4	0.05573	598.767	5894	1626	1105	1	0.23720	598.767	25067	10115	0	1
3-7-5	0.08272	614.313	8240	2969	856	1	0.17802	614.313	16847	7122	0	1
3-7-6	0.13565	650.281	12638	2471	3560	1	2.62232	650.281	200553	81010	0	1
3-7-7	0.01413	579.875	1344	320	257	1	0.08746	579.875	11097	4191	0	1
3-7-8	0.01253	621.025	1312	448	82	1	0.01701	621.025	2123	810	0	1
3-7-9	0.07163	634.735	8323	2151	1407	1	4.63934	634.735	442191	184932	0	1
3-7-10	0.00498	517.385	538	200	39	1	0.00548	517.385	632	264	0	1
3-7-11	0.22207	614.925	21808	6727	4768	1	1.22598	614.925	99089	43010	0	1
3-7-12	0.01246	665.827	1163	371	116	1	0.04329	665.827	5172	2018	0	1
3-7-13	0.00154	569.511	155	57	3	1	0.00258	569.511	201	75	0	1
3-7-14	0.00530	646.629	567	205	48	1	0.00631	646.629	685	283	0	1
3-7-15	0.48877	669.479	47211	16200	4658	1	5.03735	669.479	433362	184674	0	1
3-7-16	0.03180	583.833	3043	907	391	1	0.33041	583.833	28102	11215	0	1
3-7-17	0.00784	568.593	771	214	116	1	0.05432	568.593	5908	2360	0	1
3-7-18	0.21276	599.685	19231	5735	3482	1	0.86796	599.685	72039	30037	0	1
3-7-19	0.33289	638.693	28698	8209	5835	1	2.31804	638.693	190375	80033	0	1
3-7-20	0.19825	660.033	18574	6736	1946	1	0.73742	660.033	67441	29001	0	1
3-7-21	0.00368	532.319	384	135	35	1	0.00562	532.319	616	255	0	1
3-7-22	0.23520	652.117	23557	8851	1884	1	1.63137	652.117	144830	62002	0	1
3-7-23	0.01293	548.783	1398	547	43	1	0.01818	548.783	2174	910	0	1
3-7-24	0.00216	563.717	210	85	0	1	0.00199	563.717	210	85	0	1
3-7-25	0.00199	573.775	176	67	3	1	0.00230	573.775	233	90	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-7-26	0.12557	615.537	11894	3005	2525	1	0.90505	615.537	78914	32842	0	1
3-7-27	1.76421	649.057	150088	47865	27625	1	18.25396	649.057	1250870	537326	0	1
3-7-28	0.08448	609.743	8709	2662	1581	1	0.53906	609.743	47254	19599	0	1
3-7-29	1.34127	634.429	121266	42574	17313	1	18.79565	634.429	1368972	595831	0	1
3-7-30	0.00895	557.923	951	347	73	1	0.02046	557.923	2329	975	0	1
3-7-31	0.00292	614.925	262	94	10	1	0.00456	614.925	469	174	0	1
3-7-32	0.09063	630.165	9256	3707	607	1	0.16706	630.165	16049	7068	0	1
3-7-33	0.00472	584.445	421	127	33	1	0.02080	584.445	2376	895	0	1
3-7-34	0.00207	589.627	174	66	4	1	0.00292	589.627	291	110	0	1
3-7-35	0.01706	614.313	1607	429	216	1	0.27052	614.313	26859	10485	0	1
3-7-36	0.36186	656.381	33508	13208	2281	1	0.92795	656.381	81503	35508	0	1
3-7-37	0.08178	598.767	8553	2844	1356	1	0.32670	598.767	31454	13633	0	1
3-7-38	1.27683	644.487	104763	35691	16176	1	23.45366	644.487	1524022	660971	0	1
3-7-39	0.07067	630.471	7155	1851	1504	1	2.48282	630.471	199608	81394	0	1
3-7-40	0.34282	649.975	31782	10394	5189	1	2.25829	649.975	185190	80251	0	1
3-8-1	0.04086	690.656	3446	1270	321	1	0.07142	690.656	6171	2622	0	1
3-8-2	0.22653	727.542	18522	5877	3592	1	0.56660	727.542	43496	18771	0	1
3-8-3	0.00812	712.608	718	291	32	1	0.00963	712.608	916	394	0	1
3-8-4	0.41766	733.03	36741	12913	4579	1	2.96085	733.03	227350	99319	0	1
3-8-5	0.10010	707.12	8652	3358	682	1	0.21027	707.12	17359	7572	0	1
3-8-6	2.70561	777.832	204613	72084	33299	1	37.18762	777.832	2297058	1021076	0	1
3-8-7	0.00585	651.036	463	171	15	1	0.00752	651.036	651	264	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-8-8	0.02595	681.516	2319	856	179	1	0.10441	681.516	10282	4335	0	1
3-8-9	0.01452	686.698	1375	347	259	1	0.12942	686.698	12662	5100	0	1
3-8-10	0.01714	686.392	1504	622	50	1	0.02368	686.392	2288	989	0	1
3-8-11	0.04054	727.236	3583	1413	242	1	0.07321	727.236	6754	2958	0	1
3-8-12	0.15950	691.88	13914	5236	1281	1	1.54180	691.88	117742	51391	0	1
3-8-13	0.34116	736.988	26704	8936	4277	1	1.44080	736.988	105560	46094	0	1
3-8-14	0.01470	686.698	1453	544	96	1	0.02539	686.698	2768	1156	0	1
3-8-15	3.28069	809.23	253869	91148	26006	1	106.42383	809.23	7450664	3154664	0	1
3-8-16	0.01394	686.086	1133	432	75	1	0.02205	686.086	1895	815	0	1
3-8-17	0.02593	736.988	2063	585	344	1	0.12018	736.988	10599	4340	0	1
3-8-18	0.00937	645.854	923	378	24	1	0.01285	645.854	1324	564	0	1
3-8-19	2.12413	783.626	151587	57705	16399	1	4.84095	783.626	329583	145940	0	1
3-8-20	0.12509	700.714	11341	3772	1985	1	0.33981	700.714	27053	11638	0	1
3-8-21	0.09255	716.872	8420	2567	1203	1	3.93166	716.872	288113	119817	0	1
3-8-22	13.84514	799.784	975630	364227	114435	1	380.76617	799.784	24171515	10778833	0	1
3-8-23	0.00623	656.524	589	219	39	1	0.01803	656.524	1778	744	0	1
3-8-24	0.09383	722.36	8550	3134	772	1	2.39222	722.36	176995	77849	0	1
3-8-25	52.27156	819.9	3521203	1171940	639211	1	1000.00000	-	37123138	0	0	0
3-8-26	1.35560	737.906	101501	33110	19889	1	6.53115	737.906	432309	190374	0	1
3-8-27	1.46857	742.17	107659	37148	15029	1	10.43235	742.17	675105	293299	0	1
3-8-28	1.36900	716.566	101349	39385	10461	1	3.98654	716.566	268943	120669	0	1
3-8-29	8.37827	753.146	587894	188333	100663	1	125.36240	753.146	7681450	3348964	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
3-8-30	1.13352	722.36	87627	34355	7337	1	15.79979	722.36	1086216	482763	0	1
3-8-31	0.03754	671.764	3093	796	568	1	0.49210	671.764	46430	18858	0	1
3-8-32	6.96633	809.842	513775	180970	61855	1	261.71330	809.842	17849797	7843226	0	1
3-8-33	0.04581	759.858	3839	1259	439	1	0.15544	759.858	13309	5730	0	1
3-8-34	0.81112	747.046	64952	19970	10699	1	34.64281	747.046	2430346	1053515	0	1
3-8-35	0.03395	717.79	2782	767	354	1	0.23093	717.79	21967	8951	0	1
3-8-36	0.05252	727.236	4882	1323	775	1	0.79829	727.236	72165	29804	0	1
3-8-37	0.17099	696.756	13674	4806	1749	1	0.60223	696.756	47850	20720	0	1
3-8-38	1.99129	747.046	149212	58194	13107	1	9.31319	747.046	621522	274911	0	1
3-8-39	0.09605	721.748	8830	3595	570	1	0.22255	721.748	19799	8848	0	1
3-8-40	0.07397	695.838	6686	2593	551	1	0.22716	695.838	20035	8901	0	1
4-6-1	3.33347	853.134	238056	42487	81187	1	946.39692	853.134	58642431	23965651	0	1
4-6-2	0.89101	765.652	70601	20366	12357	1	41.24706	765.652	2618307	1094996	0	1
4-6-3	0.47682	744.312	34524	13257	2081	1	1.01538	744.312	74679	31652	0	1
4-6-4	0.77576	801.008	57292	12391	11595	1	46.46092	801.008	3230829	1232673	0	1
4-6-5	6.40566	734.254	453245	124460	96828	1	201.15198	734.254	12462611	5210770	0	1
4-6-6	0.03523	731.214	2992	789	544	1	0.17587	731.214	15124	5928	0	1
4-6-7	0.75894	785.156	70085	23920	9697	1	121.11288	785.156	8400627	3566505	0	1
4-6-8	0.13560	653.484	12901	4403	1379	1	1.98898	653.484	165948	71810	0	1
4-6-9	6.85274	785.768	487954	136521	100139	1	127.96574	785.768	7870894	3279720	0	1
4-6-10	17.79160	841.852	1202476	372473	172572	1	1000.00000	-	57283251	21415681	0	1
4-6-11	0.03942	755.594	3258	1116	240	1	0.09576	755.594	9063	3328	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-6-12	0.02745	647.69	2473	755	255	1	0.20566	647.69	18832	7645	0	1
4-6-13	0.05158	779.974	6018	2041	578	1	4.15956	779.974	345685	139688	0	1
4-6-14	0.65699	781.81	57768	20911	5144	1	4.08251	781.81	328233	137876	0	1
4-6-15	0.16997	705.304	14985	3572	3850	1	0.87519	705.304	68989	27815	0	1
4-6-16	0.04660	699.204	4488	1502	541	1	0.22113	699.204	20571	8514	0	1
4-6-17	1.92803	745.842	149172	45985	22202	1	136.71680	745.842	9163698	3909375	0	1
4-6-18	16.39220	828.448	1070534	371329	105533	1	109.89918	828.448	6695111	2794045	0	1
4-6-19	0.04840	724.502	4464	1465	525	1	1.15302	724.502	97380	42162	0	1
4-6-20	4.91408	791.868	385672	128155	59353	1	395.32232	791.868	25454123	10814512	0	1
4-6-21	2.79479	828.142	206448	65575	28647	1	28.73917	828.142	1975896	840531	0	1
4-6-22	0.56056	761.388	43732	14916	4501	1	5.52745	761.388	386918	158162	0	1
4-6-23	0.91340	714.444	80790	29107	6209	1	5.70509	714.444	460469	193668	0	1
4-6-24	1.51580	806.19	104269	23729	23002	1	40.60906	806.19	2708632	1076377	0	1
4-6-25	0.05423	705.304	4669	1070	860	1	0.23401	705.304	21721	8018	0	1
4-6-26	0.13222	736.396	11281	3469	1747	1	0.29577	736.396	24792	10121	0	1
4-6-27	3.81283	807.72	268758	85681	35837	1	83.93152	807.72	5212862	2180145	0	1
4-6-28	0.31998	746.454	27551	7215	6333	1	5.21396	746.454	349711	146377	0	1
4-6-29	4.44572	745.23	322363	101104	50416	1	74.74081	745.23	4636598	1947663	0	1
4-6-30	117.93665	874.78	7588190	2792623	768913	1	1000.00000	-	38832398	0	0	0
4-6-31	17.15113	854.052	1193081	448713	100261	1	250.72787	854.052	14928778	6397540	0	1
4-6-32	0.02191	752.554	1978	717	119	1	0.03594	752.554	3412	1402	0	1
4-6-33	45.20043	790.338	2919080	944991	389962	1	336.44529	790.338	19737081	8086023	0	1

表 B.3 – 续前页

实例 (T-S-I)	哈希去重 (开启)						哈希去重 (关闭)					
	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数	时间 (s)	解值	总扩展	剪枝数	同构剪枝	叶节点数
4-6-34	5.11267	734.866	368939	120112	48788	1	17.67519	734.866	1184730	487600	0	1
4-6-35	16.39619	822.348	1228180	256073	397670	1	1000.00000	-	39786537	0	0	0
4-6-36	2.21921	792.48	153640	46888	20811	1	16.86092	792.48	989814	396520	0	1
4-6-37	53.43447	811.678	3790434	1302849	462455	1	668.00907	811.678	42749678	17982240	0	1
4-6-38	0.09435	839.118	8165	2478	1269	1	0.31705	839.118	27203	11003	0	1
4-6-39	0.11153	756.512	10219	2432	1898	1	35.85567	756.512	2424004	954965	0	1
4-6-40	1.78928	796.744	124632	46087	8875	1	6.99591	796.744	461820	192908	0	1